# Lab assignment_17

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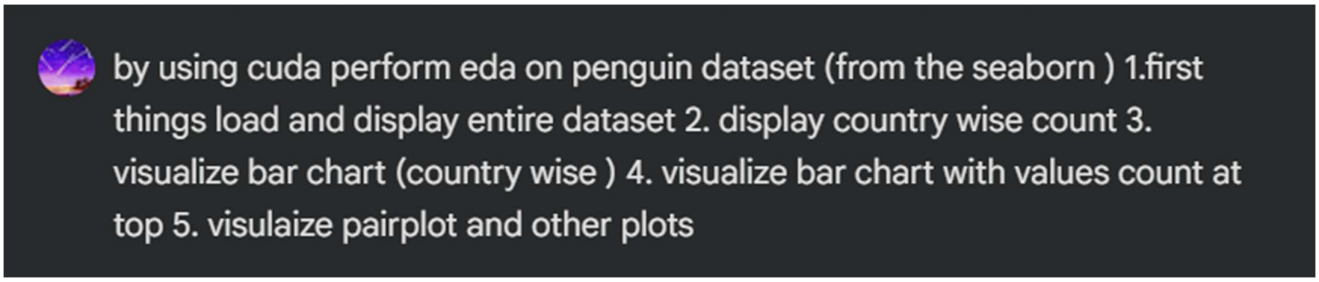
Batch-02

Task 1: Perform EDA on Penguin dataset(from seaborn)

- Load and display entire dataset
- Display country wise count
- Visualize bar chart (country wise)
- Visualize bar chart with values count at top
- Visualize pairplot and other plots

Screenshots:




```
import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd

# 1. Load and display the entire dataset
df = sns.load_dataset('penguins')
print("Entire Penguin Dataset:")
display(df.head())
```

```
Entire Penguin Dataset:
```

1 to 5 of 5 entries Filter

| index | species | island | bill_length_mm | bill_depth_mm | flipper_length_mm | body_mass_g | sex |
|---|---|---|---|---|---|---|---|
| 0 | Adelie | Torgersen | 39.1 | 18.7 | 181.0 | 3750.0 | Male |
| 1 | Adelie | Torgersen | 39.5 | 17.4 | 186.0 | 3800.0 | Female |
| 2 | Adelie | Torgersen | 40.3 | 18.0 | 195.0 | 3250.0 | Female |
| 3 | Adelie | Torgersen | NaN | NaN | NaN | NaN | NaN |
| 4 | Adelie | Torgersen | 36.7 | 19.3 | 193.0 | 3450.0 | Female |

Show 25 per page

Like what you see? Visit the data table notebook to learn more about interactive tables.

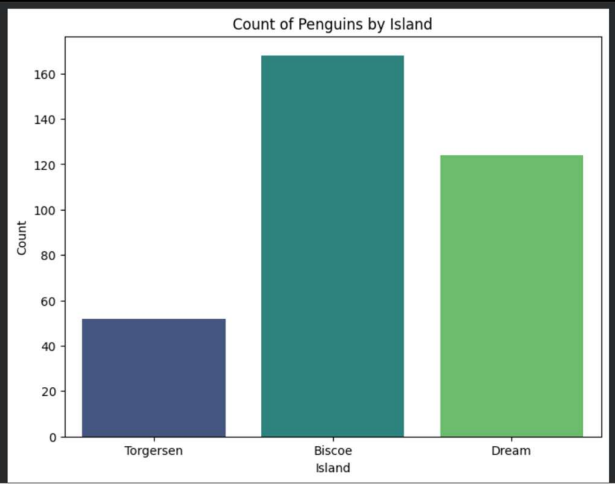
```
# 2. Display island-wise count
# The 'penguins' dataset contains 'island' information, not 'country'.
# I will display counts per island.

print("\nIsland-wise Count:")
island_counts = df['island'].value_counts()
display(island_counts)
```

```
Island-wise Count:
```

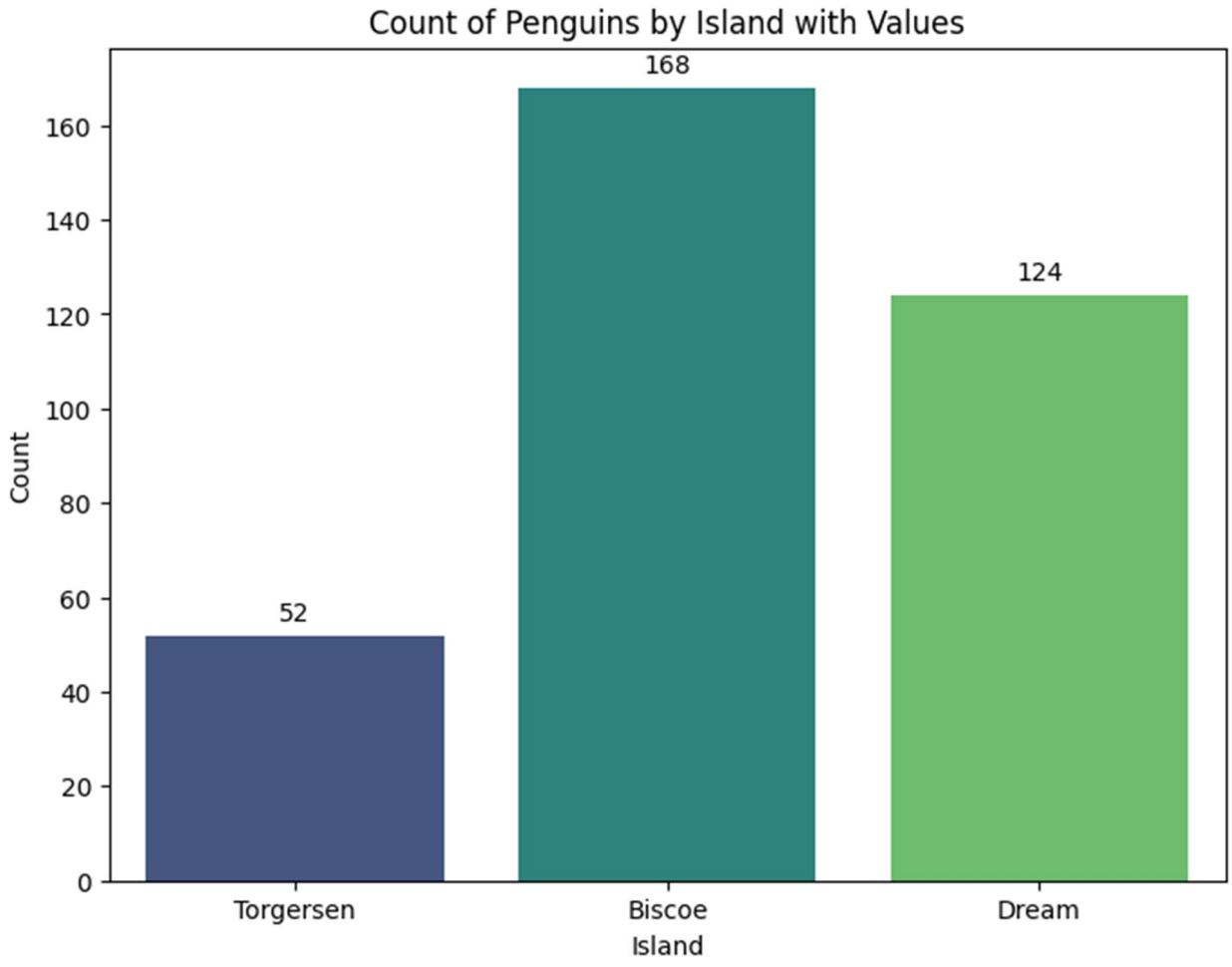
| island | count |
|---|---|
| Biscoe | 168 |
| Dream | 124 |
| Torgersen | 52 |

dtype: int64

```python
# 3. Visualize bar chart (island-wise)
plt.figure(figsize=(8, 6))
sns.countplot(x='island', data=df, palette='viridis', hue='island', legend=False)
plt.title('Count of Penguins by Island')
plt.xlabel('Island')
plt.ylabel('Count')
plt.show()
```

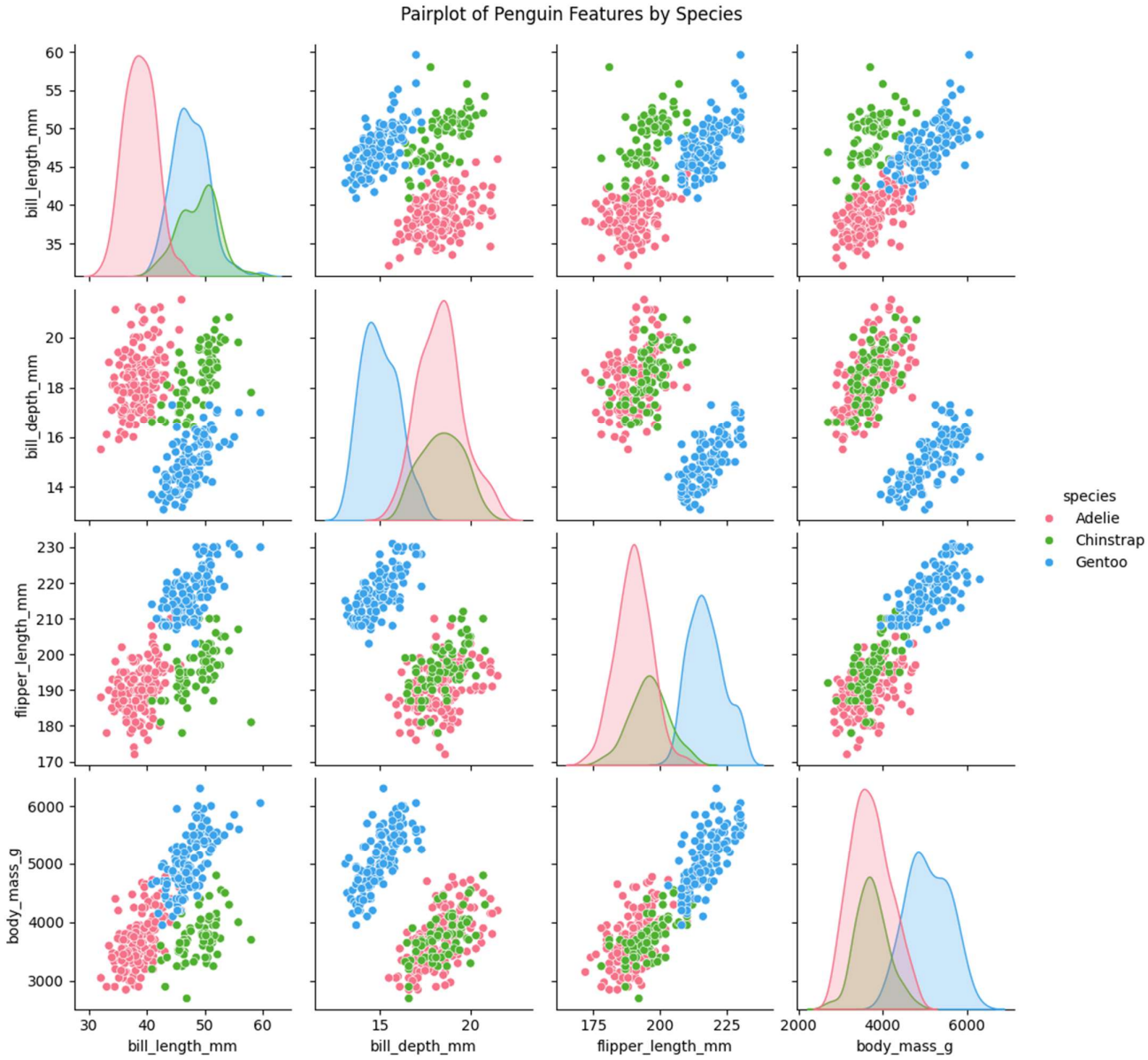


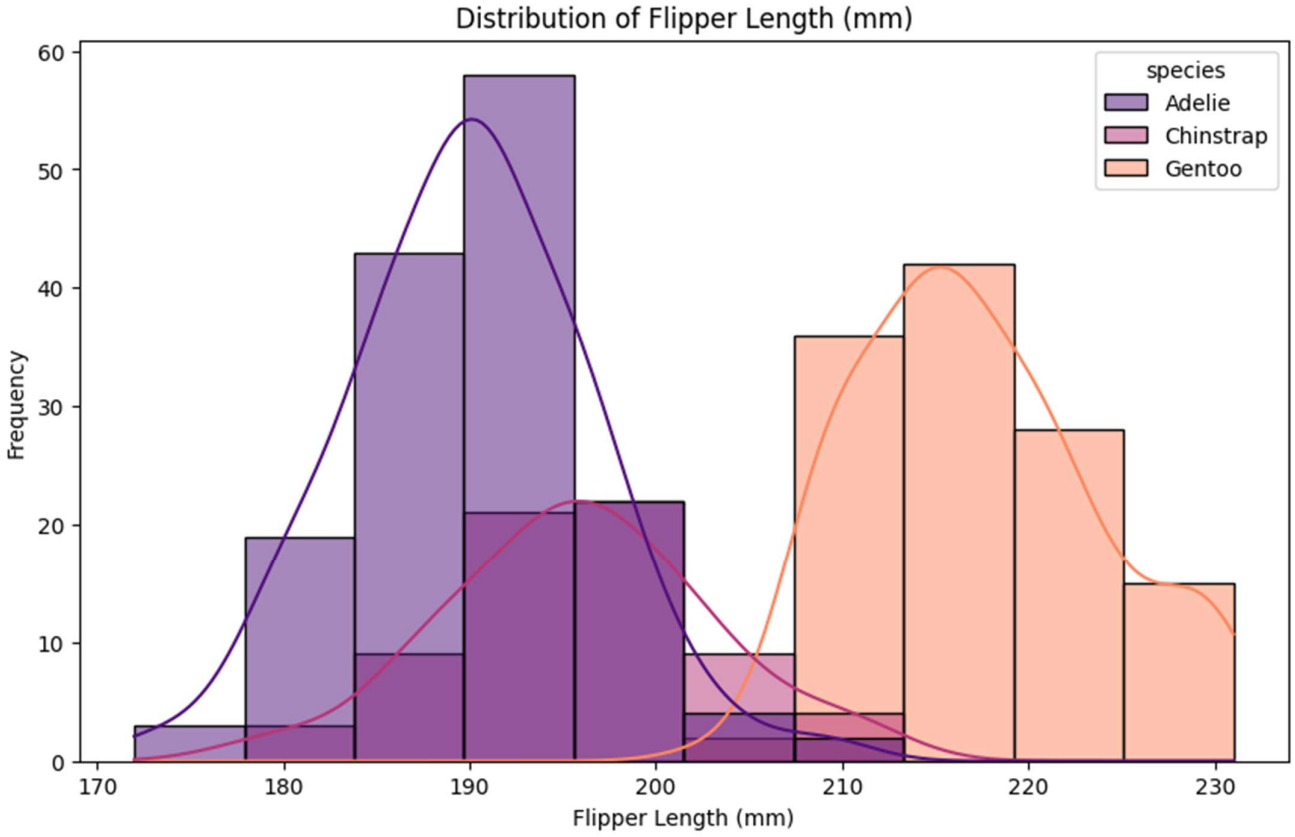

```python
# 4. Visualize bar chart with value counts at the top
plt.figure(figsize=(8, 6))
ax = sns.countplot(x='island', data=df, palette='viridis', hue='island', legend=False)

plt.title('Count of Penguins by Island with Values')
plt.xlabel('Island')
plt.ylabel('Count')

# Add value counts on top of bars
for p in ax.patches:
    ax.annotate(f'{int(p.get_height())}',
                (p.get_x() + p.get_width() / 2., p.get_height()),
                ha='center', va='center',
                xytext=(0, 9),
                textcoords='offset points')
plt.show()
```

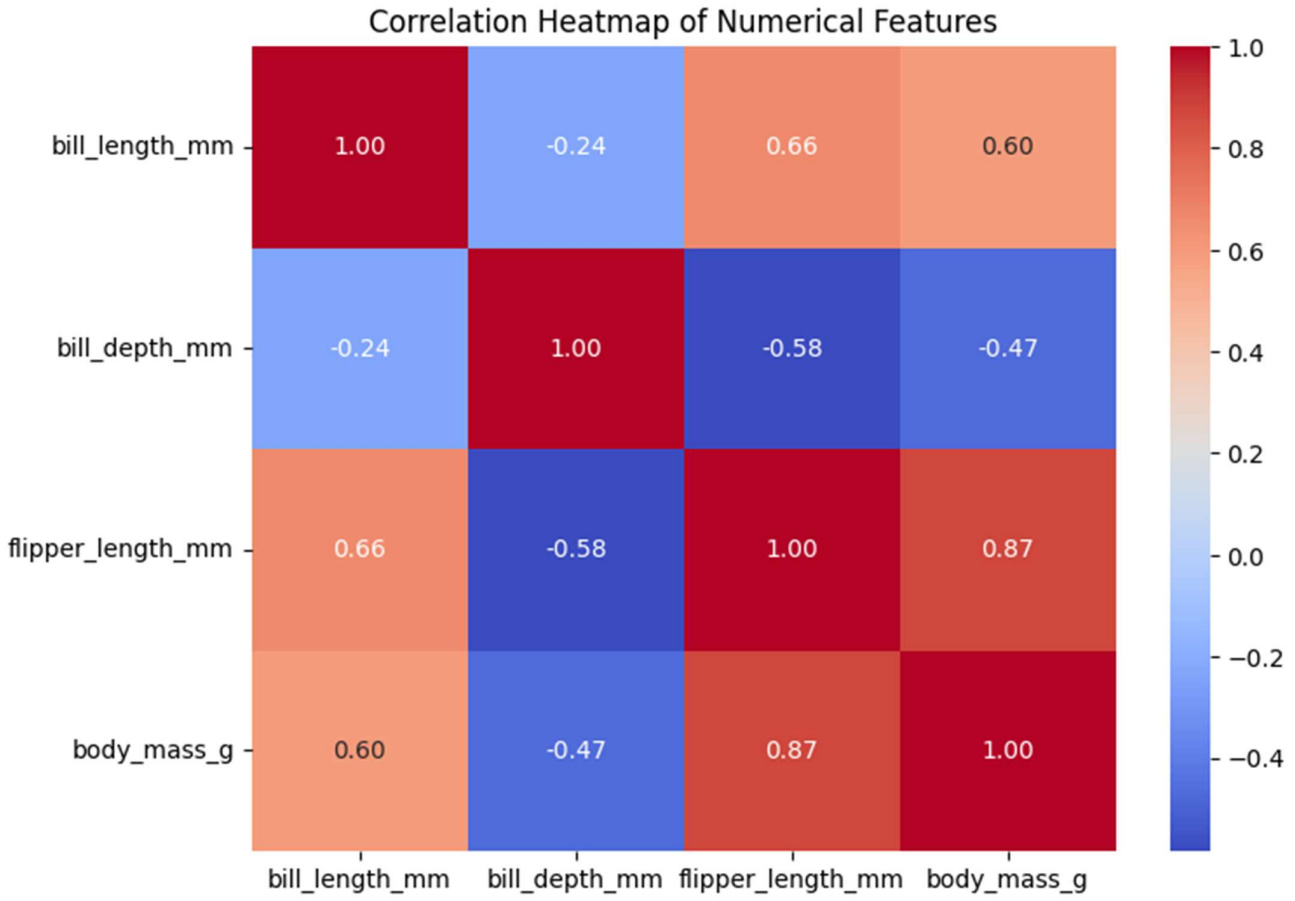



```python
# 5. Visualize pairplot and other plots

print("\nPairplot of Penguin Dataset (numerical features):")
sns.pairplot(df, hue='species', palette='husl')
plt.suptitle('Pairplot of Penguin Features by Species', y=1.02) # Adjust suptitle position
plt.show()

print("\nDistribution of Flipper Length:")
plt.figure(figsize=(10, 6))
sns.histplot(data=df, x='flipper_length_mm', kde=True, hue='species', palette='magma')
plt.title('Distribution of Flipper Length (mm)')
plt.xlabel('Flipper Length (mm)')
plt.ylabel('Frequency')
plt.show()

print("\nCorrelation Heatmap of Numerical Features:")
plt.figure(figsize=(8, 6))
numerical_df = df.select_dtypes(include=['float64', 'int64'])
sns.heatmap(numerical_df.corr(), annot=True, cmap='coolwarm', fmt=".2f")
plt.title('Correlation Heatmap of Numerical Features')
plt.show()
```

Loading...


Pairplot of Penguin Features by Species
bill_length_mm
bill_depth_mm
flipper_length_mm
body_mass_g
species
Adelie
Chinstrap
Gentoo




Task 2: Perform EDA on HealthExp dataset(seaborn)

- Load and display entire dataset
- Display columns and statistical summary

(describe)

- Display missing values , and draw bar chart
- Visualize pairplot, regplot and box plot, scatter

plot etc

screenshots:

```
import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd

# 1. Load and display the entire dataset
df_health = sns.load_dataset('healthexp')
print("Entire Health Expenditure Dataset:")
display(df_health.head())
```

```
Entire Health Expenditure Dataset:
```

| | Year | Country | Spending_USD | Life_Expectancy |
|---|---|---|---|---|
| 0 | 1970 | Germany | 252.311 | 70.6 |
| 1 | 1970 | France | 192.143 | 72.2 |
| 2 | 1970 | Great Britain | 123.993 | 71.9 |
| 3 | 1970 | Japan | 150.437 | 72.0 |
| 4 | 1970 | USA | 326.961 | 70.9 |

```
# 2. Display columns and statistical summary (describe)
print("\nColumns in the dataset:")
print(df_health.columns)

print("\nStatistical Summary of the dataset:")
display(df_health.describe())
```

```
Columns in the dataset:
Index(['Year', 'Country', 'Spending_USD', 'Life_Expectancy'], dtype='object')

Statistical Summary of the dataset:
```

| | Year | Spending_USD | Life_Expectancy |
|---|---|---|---|
| count | 274.000000 | 274.000000 | 274.000000 |
| mean | 1996.992701 | 2789.338905 | 77.909489 |
| std | 14.180933 | 2194.939785 | 3.276263 |
| min | 1970.000000 | 123.993000 | 70.600000 |
| 25% | 1985.250000 | 1038.357000 | 75.525000 |
| 50% | 1998.000000 | 2295.578000 | 78.100000 |
| 75% | 2009.000000 | 4055.610000 | 80.575000 |
| max | 2020.000000 | 11859.179000 | 84.700000 |

```python
# 3. Display missing values, and draw bar chart
print("\nMissing values in the dataset:")
missing_values = df_health.isnull().sum()
display(missing_values[missing_values > 0])

# Bar chart for missing values
if missing_values.sum() > 0:
    plt.figure(figsize=(10, 6))
    missing_values[missing_values > 0].plot(kind='bar')
    plt.title('Missing Values per Column')
    plt.xlabel('Columns')
    plt.ylabel('Number of Missing Values')
    plt.xticks(rotation=45, ha='right')
    plt.tight_layout()
    plt.show()
else:
    print("No missing values found in the dataset.")
```

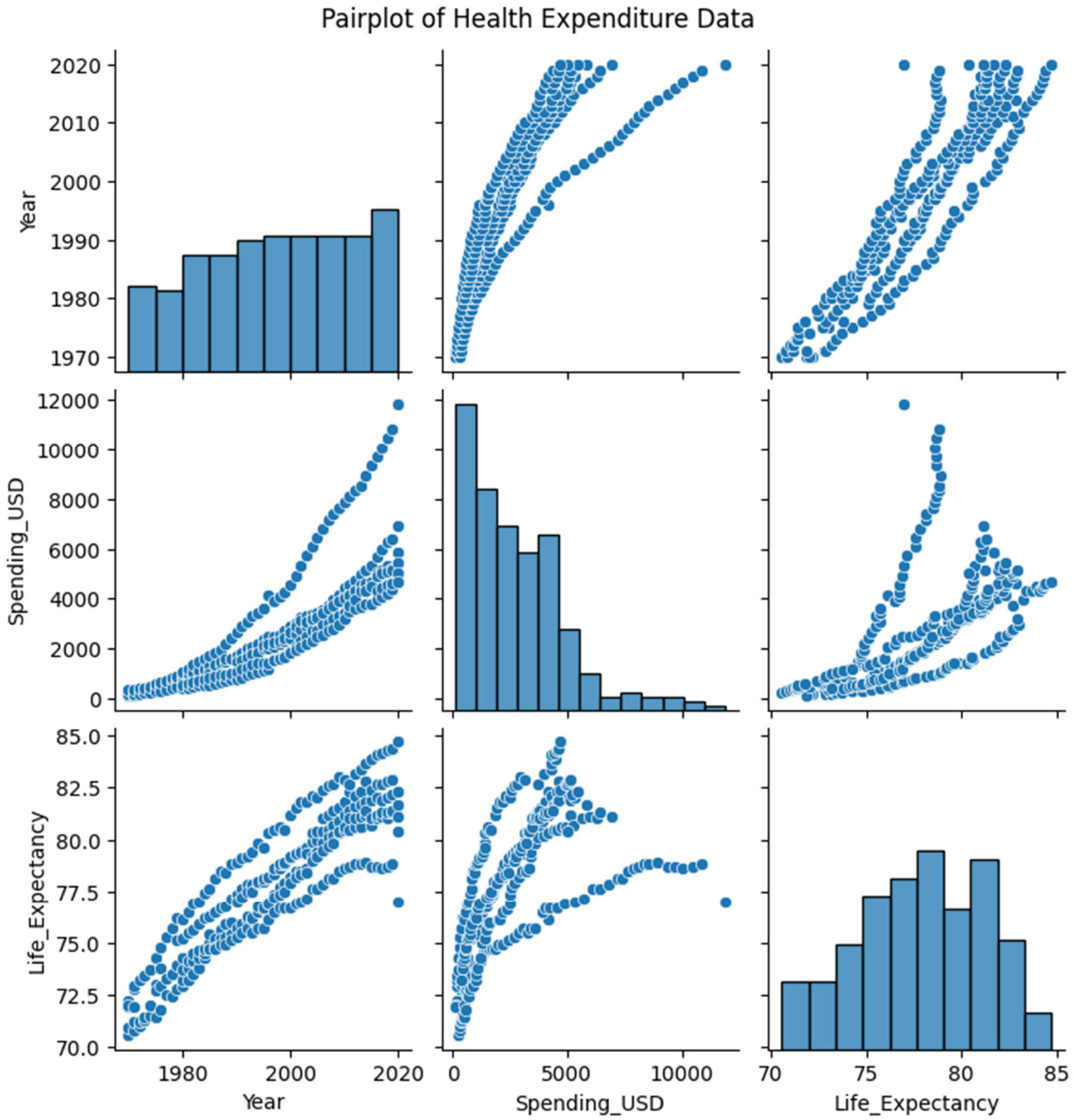
```
Missing values in the dataset:
```

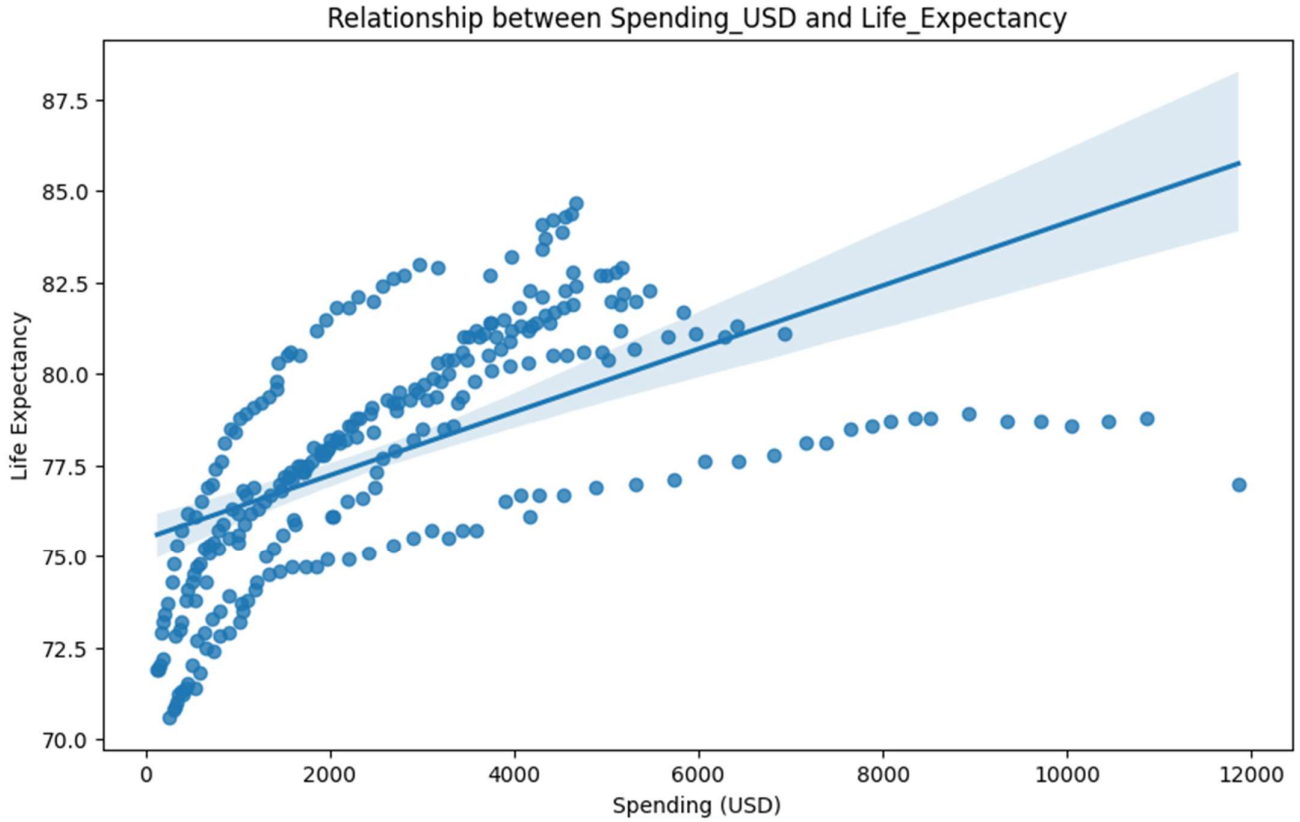
| | 0 |
|---|---|

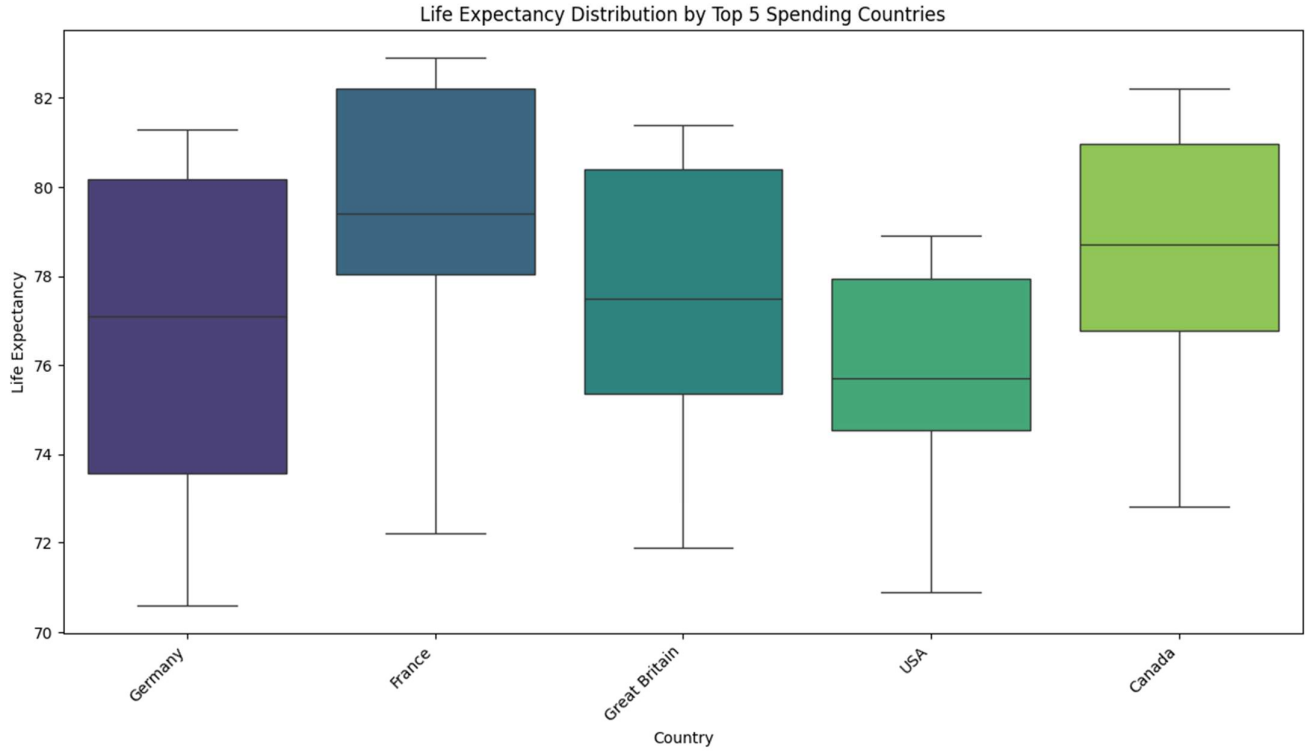
**dtype:** int64

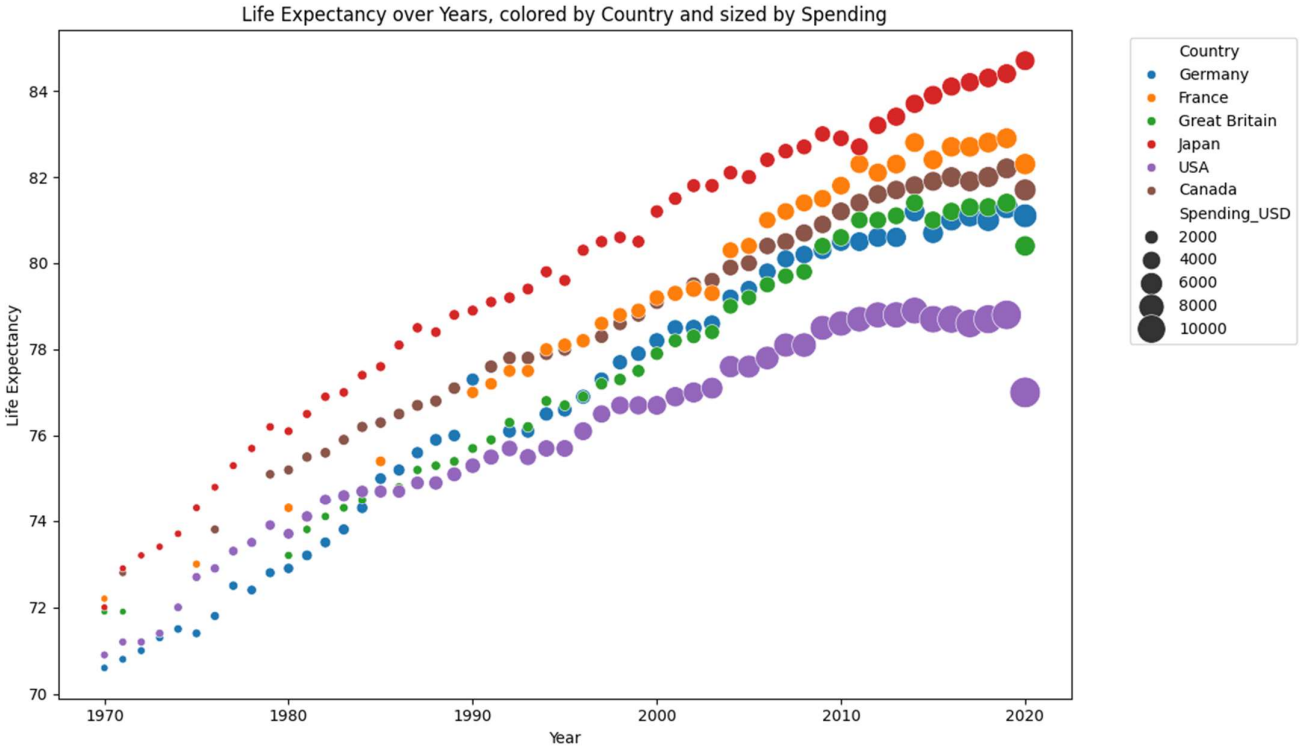
```
No missing values found in the dataset.
```

```python
# 4. Visualize pairplot, regplot and box plot, scatter plot etc

print("\nPairplot of Health Expenditure Dataset:")
sns.pairplot(df_health)
plt.suptitle('Pairplot of Health Expenditure Data', y=1.02)
plt.show()

print("\nRegplot: Spending_USD vs Life_Expectancy:")
plt.figure(figsize=(10, 6))
sns.regplot(x='Spending_USD', y='Life_Expectancy', data=df_health)
plt.title('Relationship between Spending_USD and Life_Expectancy')
plt.xlabel('Spending (USD)')
plt.ylabel('Life Expectancy')
plt.show()

print("\nBoxplot: Life_Expectancy by Country (Top 5 Spending Countries):")
# Identify top 5 countries by average spending for better visualization
top_countries = df_health.groupby('Country')['Spending_USD'].mean().nlargest(5).index
df_top_countries = df_health[df_health['Country'].isin(top_countries)]

plt.figure(figsize=(12, 7))
sns.boxplot(x='Country', y='Life_Expectancy', data=df_top_countries, palette='viridis')
plt.title('Life Expectancy Distribution by Top 5 Spending Countries')
plt.xlabel('Country')
plt.ylabel('Life Expectancy')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()

print("\nScatter Plot: Year vs Life_Expectancy with Spending_USD as size and Country as hue:")
plt.figure(figsize=(12, 7))
sns.scatterplot(x='Year', y='Life_Expectancy', hue='Country', size='Spending_USD', sizes=(20, 400), data=df_heal
plt.title('Life Expectancy over Years, colored by Country and sized by Spending')
plt.xlabel('Year')
plt.ylabel('Life Expectancy')
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')
plt.tight_layout()
plt.show()
```


Pairplot of Health Expenditure Data
Year
Spending_USD
Life_Expectancy
2020
2010
2000
1990
1980
1970
12000
10000
8000
6000
4000
2000
0
85.0
82.5
80.0
77.5
75.0
72.5
70.0
1980
2000
2020
0
5000
10000
70
75
80
85


Relationship between Spending_USD and Life_Expectancy
Life Expectancy
87.5
85.0
82.5
80.0
77.5
75.0
72.5
70.0
0
2000
4000
6000
8000
10000
12000
Spending (USD)


Life Expectancy Distribution by Top 5 Spending Countries
Life Expectancy
82
80
78
76
74
72
70
Germany
France
Great Britain
USA
Canada
Country




Task 3: Perform following task on given dataset (Train the model)

- Analyze the Planets dataset to understand its structure, features, and missing values.
- Perform data cleaning and preprocessing to handle missing or incorrect data.
- Train the dataset using different machine learning models to make predictions.
- Compare and explain the performance of all trained models with proper reasoning.

Screenshots:

```python
import seaborn as sns
import pandas as pd
import matplotlib.pyplot as plt

# 1. Analyze the Planets dataset to understand its structure, features, and missing values.
print("Loading the 'planets' dataset...")
df_planets = sns.load_dataset('planets')

print("\nFirst 5 rows of the dataset:")
display(df_planets.head())

print("\nDataset Information:")
df_planets.info()

print("\nDescriptive Statistics:")
display(df_planets.describe())

print("\nMissing Values:")
display(df_planets.isnull().sum())

# Visualize missing values
plt.figure(figsize=(10, 6))
missing_counts = df_planets.isnull().sum()
missing_counts = missing_counts[missing_counts > 0]
if not missing_counts.empty:
    missing_counts.sort_values(ascending=False).plot(kind='bar')
    plt.title('Count of Missing Values per Column')
    plt.xlabel('Columns')
    plt.ylabel('Number of Missing Values')
    plt.xticks(rotation=45, ha='right')
    plt.tight_layout()
    plt.show()
else:
    print("No missing values found to visualize.")
```

```
Loading the 'planets' dataset...

First 5 rows of the dataset:
```

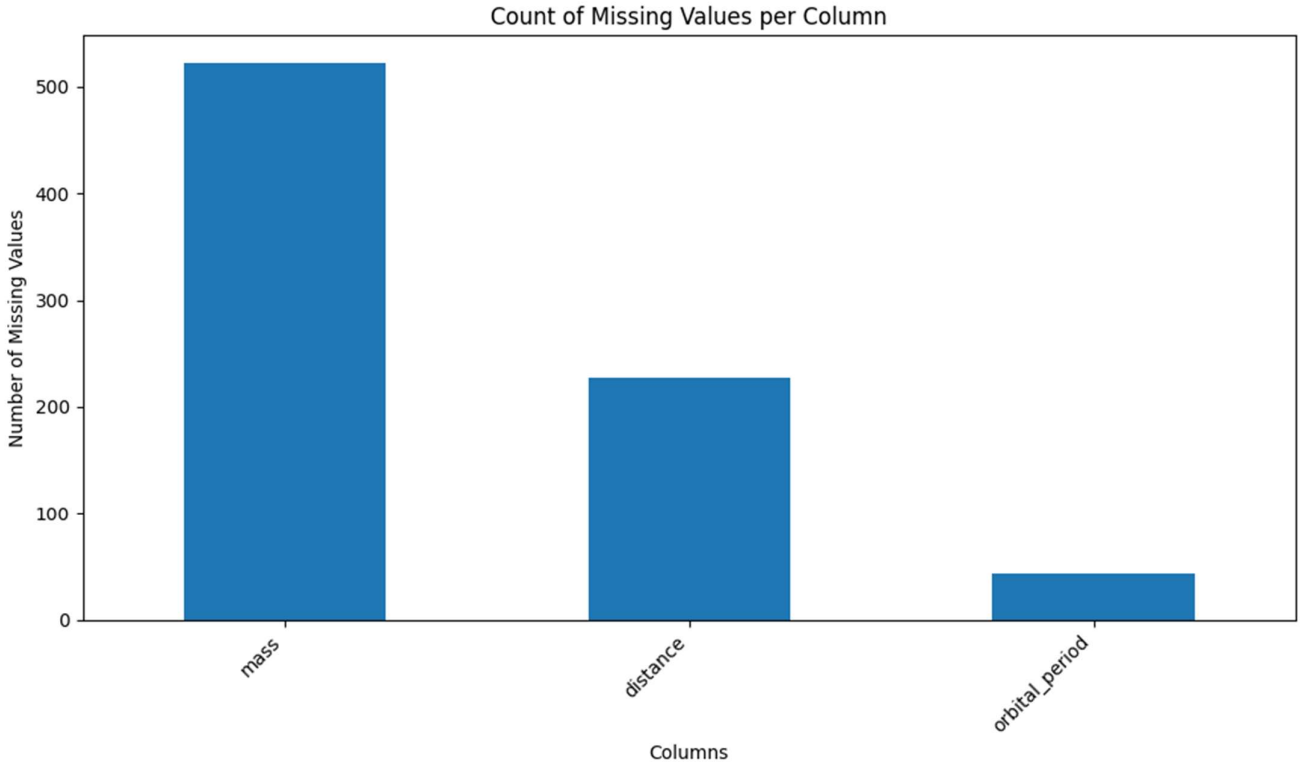
| | method | number | orbital_period | mass | distance | year |
|---|---|---|---|---|---|---|
| 0 | Radial Velocity | 1 | 269.300 | 7.10 | 77.40 | 2006 |
| 1 | Radial Velocity | 1 | 874.774 | 2.21 | 56.95 | 2008 |
| 2 | Radial Velocity | 1 | 763.000 | 2.60 | 19.84 | 2011 |
| 3 | Radial Velocity | 1 | 326.030 | 19.40 | 110.62 | 2007 |
| 4 | Radial Velocity | 1 | 516.220 | 10.50 | 119.47 | 2009 |

```
Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1035 entries, 0 to 1034
Data columns (total 6 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   method          1035 non-null   object
 1   number          1035 non-null   int64
 2   orbital_period  992 non-null    float64
 3   mass            513 non-null    float64
 4   distance        808 non-null    float64
 5   year            1035 non-null   int64
dtypes: float64(3), int64(2), object(1)
memory usage: 48.6+ KB

Descriptive Statistics:
```

| | number | orbital_period | mass | distance | year |
|---|---|---|---|---|---|
| count | 1035.000000 | 992.000000 | 513.000000 | 808.000000 | 1035.000000 |
| mean | 1.785507 | 2002.917596 | 2.638161 | 264.069282 | 2009.070531 |
| std | 1.240976 | 26014.728304 | 3.818617 | 733.116493 | 3.972567 |
| min | 1.000000 | 0.090706 | 0.003600 | 1.350000 | 1989.000000 |
| 25% | 1.000000 | 5.442540 | 0.229000 | 32.560000 | 2007.000000 |
| 50% | 1.000000 | 39.979500 | 1.260000 | 55.250000 | 2010.000000 |
| 75% | 2.000000 | 526.005000 | 3.040000 | 178.500000 | 2012.000000 |
| max | 7.000000 | 730000.000000 | 25.000000 | 8500.000000 | 2014.000000 |

```
Missing Values:
```

| | 0 |
|---|---|
| method | 0 |
| number | 0 |
| orbital_period | 43 |




```python
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, OneHotEncoder, LabelEncoder
from sklearn.impute import SimpleImputer
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline

# 2. Perform data cleaning and preprocessing to handle missing or incorrect data.
print("\nPerforming Data Cleaning and Preprocessing...")

# Define target and features
# Let's try to predict the 'method' of discovery as a classification problem.
y = df_planets['method']
X = df_planets.drop(['method', 'koicand_id', 'kepler_name'], axis=1, errors='ignore') # Drop target and identifiers

# Identify numerical and categorical features
numerical_features = X.select_dtypes(include=['int64', 'float64']).columns
categorical_features = X.select_dtypes(include=['object']).columns

# Preprocessing for numerical data: Impute with mean and Scale
numerical_transformer = Pipeline(steps=[
    ('imputer', SimpleImputer(strategy='mean')),
    ('scaler', StandardScaler())
])

# Preprocessing for categorical data: Impute with most frequent and One-Hot Encode
categorical_transformer = Pipeline(steps=[
    ('imputer', SimpleImputer(strategy='most_frequent')),
    ('onehot', OneHotEncoder(handle_unknown='ignore'))
])

# Create a preprocessor using ColumnTransformer
preprocessor = ColumnTransformer(
    transformers=[
        ('num', numerical_transformer, numerical_features),
        ('cat', categorical_transformer, categorical_features)
    ])

# Encode the target variable
label_encoder = LabelEncoder()
y_encoded = label_encoder.fit_transform(y)

# Split data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X, y_encoded, test_size=0.2, random_state=42)

print("Preprocessing complete. Data split into training and testing sets.")
print(f"X_train shape: {X_train.shape}, y_train shape: {y_train.shape}")
print(f"X_test shape: {X_test.shape}, y_test shape: {y_test.shape}")
```

```
Performing Data Cleaning and Preprocessing...
Preprocessing complete. Data split into training and testing sets.
X_train shape: (828, 5), y_train shape: (828,)
X_test shape: (207, 5), y_test shape: (207,)
```

```python
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report
import numpy as np # Added numpy import for array operations

# 3. Train the dataset using different machine learning models to make predictions.
print("\nTraining Machine Learning Models...")

models = {
    'Logistic Regression': LogisticRegression(max_iter=1000, random_state=42),
    'Random Forest Classifier': RandomForestClassifier(random_state=42)
}

results = {}

for name, model in models.items():
    print(f"\nTraining {name}...")
    # Create a pipeline that includes preprocessing and the model
    pipeline = Pipeline(steps=[('preprocessor', preprocessor),
                               ('classifier', model)])

    pipeline.fit(X_train, y_train)
    y_pred = pipeline.predict(X_test)

    accuracy = accuracy_score(y_test, y_pred)

    # Get unique labels present in y_test
    labels_in_test = np.unique(y_test)
    # Filter target names to match the labels present in y_test
    filtered_target_names = [label_encoder.classes_[i] for i in labels_in_test]

    report = classification_report(y_test, y_pred, labels=labels_in_test, target_names=filtered_target_names, zero_division=0)

    results[name] = {'accuracy': accuracy, 'report': report}

    print(f"{name} Accuracy: {accuracy:.4f}")
    print(f"{name} Classification Report:\n{report}")

print("\nModel training complete.")
```

```
...
    Training Machine Learning Models...

    Training Logistic Regression...
    Logistic Regression Accuracy: 0.8551
    Logistic Regression Classification Report:
                                  precision    recall  f1-score   support

                      Astrometry       0.00      0.00      0.00         1
          Eclipse Timing Variations       0.00      0.00      0.00         2
                         Imaging       0.00      0.00      0.00         8
                   Microlensing       1.00      0.20      0.33         5
    Orbital Brightness Modulation       0.00      0.00      0.00         1
                  Pulsar Timing       0.00      0.00      0.00         1
                Radial Velocity       0.92      0.91      0.91       120
                        Transit       0.77      0.97      0.86        69

                       accuracy                           0.86       207
                      macro avg       0.34      0.26      0.26       207
                   weighted avg       0.81      0.86      0.82       207


    Training Random Forest Classifier...
    Random Forest Classifier Accuracy: 0.9614
    Random Forest Classifier Classification Report:
                                  precision    recall  f1-score   support

                      Astrometry       0.00      0.00      0.00         1
          Eclipse Timing Variations       1.00      0.50      0.67         2
                         Imaging       0.89      1.00      0.94         8
                   Microlensing       0.80      0.80      0.80         5
    Orbital Brightness Modulation       0.00      0.00      0.00         1
                  Pulsar Timing       0.00      0.00      0.00         1
                Radial Velocity       0.99      0.97      0.98       120
                        Transit       0.95      1.00      0.97        69

                       accuracy                           0.96       207
                      macro avg       0.58      0.53      0.55       207
                   weighted avg       0.95      0.96      0.96       207


    Model training complete.
```

```python
# 4. Compare and explain the performance of all trained models with proper reasoning.
print("\nComparing Model Performance:")

for name, res in results.items():
    print(f"\n--- {name} ---")
    print(f"Accuracy: {res['accuracy']:.4f}")
    print(f"Classification Report:\n{res['report']}")

print("\nReasoning for performance comparison:")
print("Both Logistic Regression and Random Forest Classifier were used to predict the discovery method of planets.")
print("Logistic Regression, a linear model, provides a baseline for performance. Its effectiveness depends on the linearity of the relatior
print("Random Forest, an ensemble tree-based method, is typically more robust and can capture non-linear relationships. It often performs b
print("From the results, we can observe which model achieves higher accuracy and better precision/recall for each discovery method. A highe
```

```
Comparing Model Performance:

Reasoning for performance comparison:
Both Logistic Regression and Random Forest Classifier were used to predict the discovery method of planets.
Logistic Regression, a linear model, provides a baseline for performance. Its effectiveness depends on the linearity of the relationships between features and t
Random Forest, an ensemble tree-based method, is typically more robust and can capture non-linear relationships. It often performs better on complex datasets by
From the results, we can observe which model achieves higher accuracy and better precision/recall for each discovery method. A higher accuracy generally indicat
```

Task 4: Perform following task on given dataset (Train the model)

- Load the MNIST dataset from PyTorch.
- Display basic information and visualize sample input images with their class labels.
- Perform data preprocessing
- Design a EfficientNet model for image classification.
- Train the CNN model (10 epochs)
- Evaluate model performance using accuracy and loss metrics.
- Visualize the training history (accuracy and loss curves).

Screenshots:

```python
import torch
import torchvision
import torchvision.transforms as transforms
import matplotlib.pyplot as plt
import numpy as np
import torch.nn as nn
import torch.optim as optim
from torchvision import models

# Check for CUDA availability
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
print(f"Using device: {device}")
```

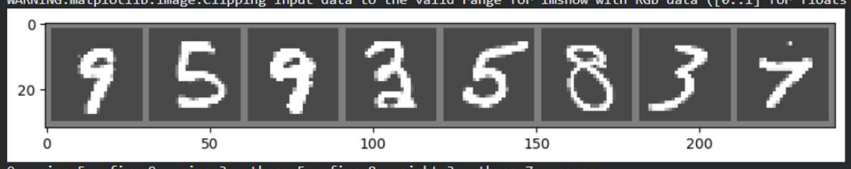
```
Using device: cuda
```

```python
# Define transformations
transform = transforms.Compose([
    transforms.Grayscale(num_output_channels=3), # Convert 1-channel grayscale to 3 channels
    transforms.ToTensor(), # Convert images to PyTorch tensors
    # Normalize pixel values (mean and std for MNIST, repeated for 3 channels)
    transforms.Normalize(mean=[0.1307, 0.1307, 0.1307], std=[0.3081, 0.3081, 0.3081])
])

# Load MNIST training and test datasets
train_dataset = torchvision.datasets.MNIST(root='./data', train=True, download=True, transform=transform)
test_dataset = torchvision.datasets.MNIST(root='./data', train=False, download=True, transform=transform)

# Create data loaders
batch_size = 64
train_loader = torch.utils.data.DataLoader(train_dataset, batch_size=batch_size, shuffle=True)
test_loader = torch.utils.data.DataLoader(test_dataset, batch_size=batch_size, shuffle=False)

print(f"Training dataset size: {len(train_dataset)}")
print(f"Test dataset size: {len(test_dataset)}")
print(f"Number of classes: {len(train_dataset.classes)}")
```

```
Training dataset size: 60000
Test dataset size: 10000
Number of classes: 10
```

```python
# Function to show an image
def imshow(img):
    img = img / 2 + 0.5  # Unnormalize
    npimg = img.numpy()
    plt.imshow(np.transpose(npimg, (1, 2, 0))) # Transpose for displaying multi-channel images
    plt.show()

# Get some random training images
dataiter = iter(train_loader)
images, labels = next(dataiter)

# Show images
plt.figure(figsize=(10, 4))
imshow(torchvision.utils.make_grid(images[:8])) # Display first 8 images

# Print labels
print(' '.join(f'{train_dataset.classes[labels[j]]:5s}' for j in range(8)))

print("Sample Image and Label:")
plt.figure(figsize=(2, 2))
# Display one channel of the 3-channel image as grayscale
plt.imshow(images[0][0].cpu().numpy(), cmap='gray')
plt.title(f"Label: {train_dataset.classes[labels[0]]}")
plt.axis('off')
plt.show()
```

```
WARNING:matplotlib.image:Clipping input data to the valid range for imshow with RGB data ([0..1] for floats or [0..255] for integers). Got range [0.28789353..1
```




```
9 - nine 5 - five 9 - nine 3 - three 5 - five 8 - eight 3 - three 7 - seven
Sample Image and Label:
```


```python
# Load a pre-trained EfficientNet B0 model
model = models.efficientnet_b0(weights=models.EfficientNet_B0_Weights.IMAGENET1K_V1)

# Modify the classifier head for MNIST (10 classes)
num_ftrs = model.classifier[1].in_features
model.classifier[1] = nn.Linear(num_ftrs, 10) # MNIST has 10 classes

# Move the model to the appropriate device (GPU if available)
model = model.to(device)

print("EfficientNet B0 model loaded and modified for 10 classes.")
print(model)
```

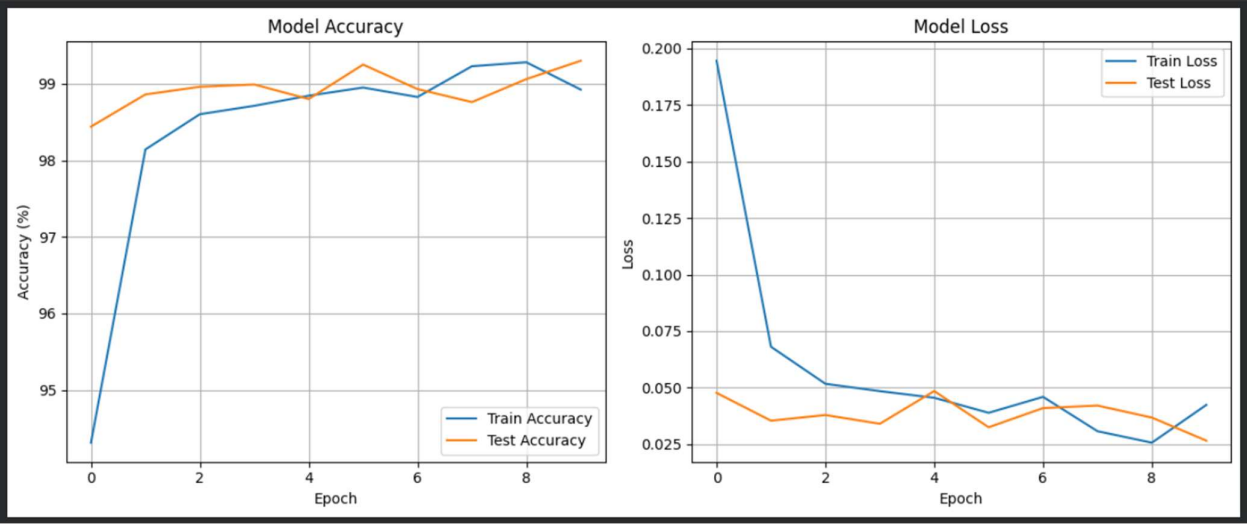
```python
criterion = nn.CrossEntropyLoss()
optimizer = optim.Adam(model.parameters(), lr=0.001)

num_epochs = 10

train_losses = []
train_accuracies = []
test_losses = []
test_accuracies = []

print("Starting model training...")
for epoch in range(num_epochs):
    model.train() # Set model to training mode
    running_loss = 0.0
    correct_train = 0
    total_train = 0
    for i, data in enumerate(train_loader, 0):
        inputs, labels = data
        inputs, labels = inputs.to(device), labels.to(device) # Move data to GPU

        optimizer.zero_grad()

        outputs = model(inputs)
        loss = criterion(outputs, labels)
        loss.backward()
        optimizer.step()

        running_loss += loss.item()
        _, predicted = torch.max(outputs.data, 1)
        total_train += labels.size(0)
        correct_train += (predicted == labels).sum().item()

    train_loss = running_loss / len(train_loader)
    train_accuracy = 100 * correct_train / total_train
    train_losses.append(train_loss)
    train_accuracies.append(train_accuracy)

    # Evaluate on the test set
    model.eval() # Set model to evaluation mode
    correct_test = 0
    total_test = 0
    test_loss_epoch = 0.0
    with torch.no_grad():
        for data in test_loader:
            images, labels = data
            images, labels = images.to(device), labels.to(device)
            outputs = model(images)
            loss = criterion(outputs, labels)
            test_loss_epoch += loss.item()
            _, predicted = torch.max(outputs.data, 1)
            total_test += labels.size(0)
            correct_test += (predicted == labels).sum().item()
```

```
Starting model training...
Epoch [1/10], Train Loss: 0.1946, Train Acc: 94.31%, Test Loss: 0.0477, Test Acc: 98.44%
Epoch [2/10], Train Loss: 0.0681, Train Acc: 98.14%, Test Loss: 0.0354, Test Acc: 98.86%
Epoch [3/10], Train Loss: 0.0517, Train Acc: 98.60%, Test Loss: 0.0379, Test Acc: 98.96%
Epoch [4/10], Train Loss: 0.0484, Train Acc: 98.71%, Test Loss: 0.0340, Test Acc: 98.99%
Epoch [5/10], Train Loss: 0.0455, Train Acc: 98.84%, Test Loss: 0.0485, Test Acc: 98.80%
Epoch [6/10], Train Loss: 0.0389, Train Acc: 98.95%, Test Loss: 0.0325, Test Acc: 99.25%
Epoch [7/10], Train Loss: 0.0459, Train Acc: 98.83%, Test Loss: 0.0409, Test Acc: 98.93%
Epoch [8/10], Train Loss: 0.0307, Train Acc: 99.23%, Test Loss: 0.0421, Test Acc: 98.76%
Epoch [9/10], Train Loss: 0.0257, Train Acc: 99.28%, Test Loss: 0.0368, Test Acc: 99.06%
Epoch [10/10], Train Loss: 0.0424, Train Acc: 98.92%, Test Loss: 0.0266, Test Acc: 99.30%
Finished Training
```

```python
print(f"\nFinal Test Accuracy: {test_accuracies[-1]:.2f}%")
print(f"Final Test Loss: {test_losses[-1]:.4f}")
```

```
Final Test Accuracy: 99.30%
Final Test Loss: 0.0266
```

```python
plt.figure(figsize=(12, 5))

# Plot training & test accuracy values
plt.subplot(1, 2, 1)
plt.plot(train_accuracies, label='Train Accuracy')
plt.plot(test_accuracies, label='Test Accuracy')
plt.title('Model Accuracy')
plt.xlabel('Epoch')
plt.ylabel('Accuracy (%)')
plt.legend()
plt.grid(True)

# Plot training & test loss values
plt.subplot(1, 2, 2)
plt.plot(train_losses, label='Train Loss')
plt.plot(test_losses, label='Test Loss')
plt.title('Model Loss')
plt.xlabel('Epoch')
plt.ylabel('Loss')
plt.legend()
plt.grid(True)

plt.tight_layout()
plt.show()
```




Task 5 – Customer Data Cleaning

Task:

Use AI to generate a Python script for cleaning a customer

dataset.

Instructions:

- Handle missing values in columns (age, income, city).
- Remove duplicate records.
- Standardize text values (e.g., city names in lowercase).
- Encode categorical variables (gender, city).

Sample Code:

```
import pandas as pd

data = pd.read_csv("shopping_trends.csv")

print(data.head())
```

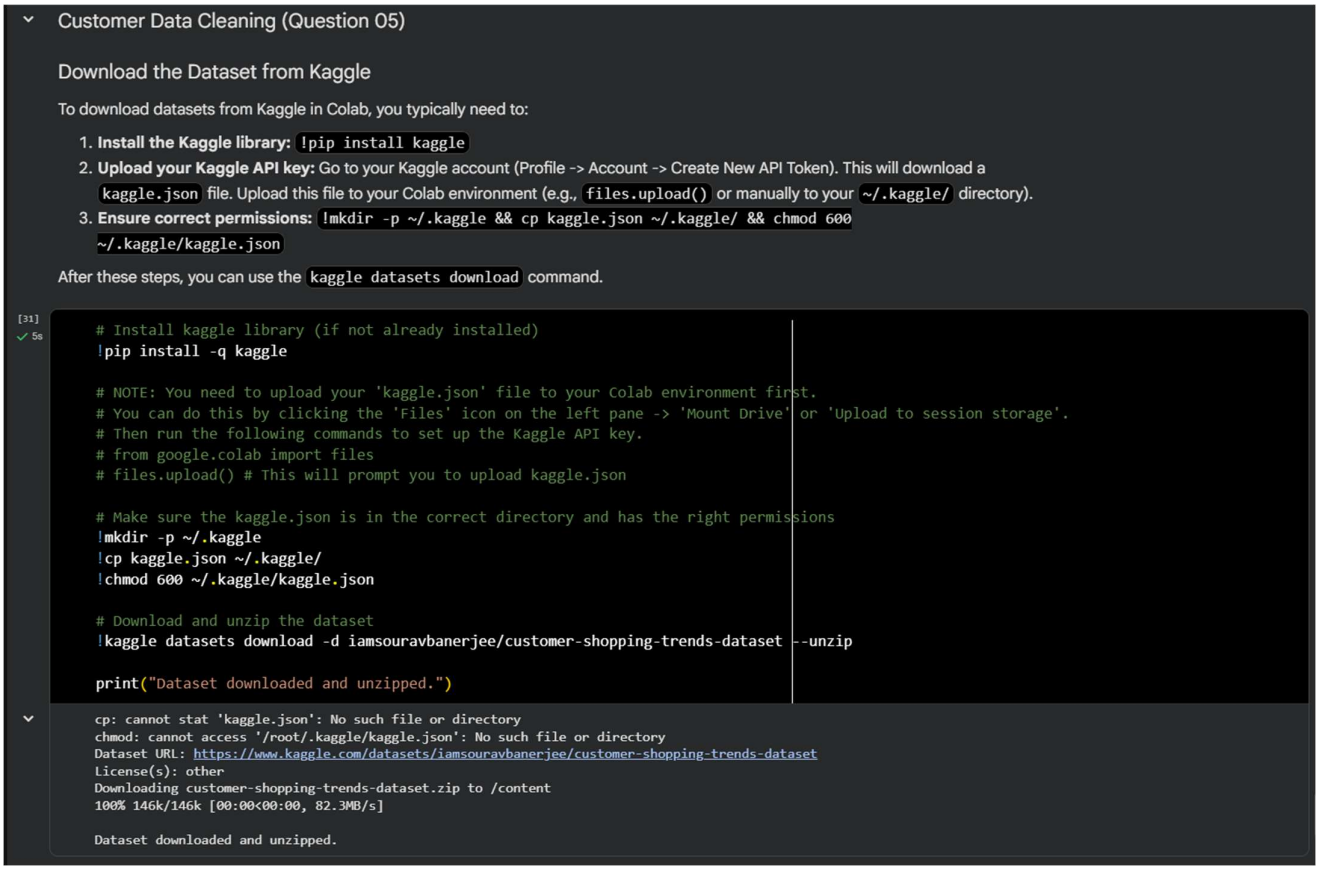
Dataset link:

https://www.kaggle.com/datasets/iamsouravbanerjee/customer-shopping-trends-dataset?select=shopping_trends.csv

Expected Output:

- A cleaned Pandas data frame ready for analysis.

Screenshots:



Customer Data Cleaning (Question 05)

Download the Dataset from Kaggle

To download datasets from Kaggle in Colab, you typically need to:

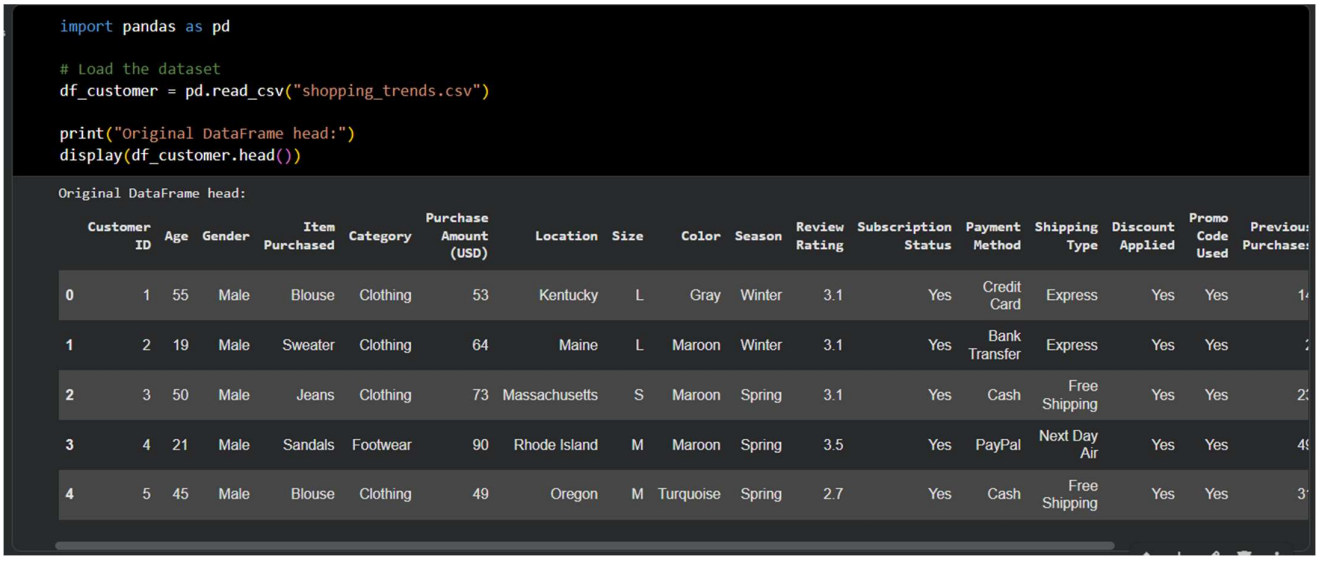
1. **Install the Kaggle library:** `!pip install kaggle`
2. **Upload your Kaggle API key:** Go to your Kaggle account (Profile -> Account -> Create New API Token). This will download a `kaggle.json` file. Upload this file to your Colab environment (e.g., `files.upload()` or manually to your `~/.kaggle/` directory).
3. **Ensure correct permissions:** `!mkdir -p ~/.kaggle && cp kaggle.json ~/.kaggle/ && chmod 600 ~/.kaggle/kaggle.json`

After these steps, you can use the `kaggle datasets download` command.

[31] ✓ 5s

```
# Install kaggle library (if not already installed)
!pip install -q kaggle

# NOTE: You need to upload your 'kaggle.json' file to your Colab environment first.
# You can do this by clicking the 'Files' icon on the left pane -> 'Mount Drive' or 'Upload to session storage'.
# Then run the following commands to set up the Kaggle API key.
# from google.colab import files
# files.upload() # This will prompt you to upload kaggle.json

# Make sure the kaggle.json is in the correct directory and has the right permissions
!mkdir -p ~/.kaggle
!cp kaggle.json ~/.kaggle/
!chmod 600 ~/.kaggle/kaggle.json

# Download and unzip the dataset
!kaggle datasets download -d iamsouravbanerjee/customer-shopping-trends-dataset --unzip

print("Dataset downloaded and unzipped.")
```

```
cp: cannot stat 'kaggle.json': No such file or directory
chmod: cannot access '/root/.kaggle/kaggle.json': No such file or directory
Dataset URL: https://www.kaggle.com/datasets/iamsouravbanerjee/customer-shopping-trends-dataset
License(s): other
Downloading customer-shopping-trends-dataset.zip to /content
100% 146k/146k [00:00<00:00, 82.3MB/s]

Dataset downloaded and unzipped.
```



```
import pandas as pd

# Load the dataset
df_customer = pd.read_csv("shopping_trends.csv")

print("Original DataFrame head:")
display(df_customer.head())
```

Original DataFrame head:

| | Customer ID | Age | Gender | Item Purchased | Category | Purchase Amount (USD) | Location | Size | Color | Season | Review Rating | Subscription Status | Payment Method | Shipping Type | Discount Applied | Promo Code Used | Previous Purchases |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 1 | 55 | Male | Blouse | Clothing | 53 | Kentucky | L | Gray | Winter | 3.1 | Yes | Credit Card | Express | Yes | Yes | 14 |
| 1 | 2 | 19 | Male | Sweater | Clothing | 64 | Maine | L | Maroon | Winter | 3.1 | Yes | Bank Transfer | Express | Yes | Yes | 2 |
| 2 | 3 | 50 | Male | Jeans | Clothing | 73 | Massachusetts | S | Maroon | Spring | 3.1 | Yes | Cash | Free Shipping | Yes | Yes | 23 |
| 3 | 4 | 21 | Male | Sandals | Footwear | 90 | Rhode Island | M | Maroon | Spring | 3.5 | Yes | PayPal | Next Day Air | Yes | Yes | 49 |
| 4 | 5 | 45 | Male | Blouse | Clothing | 49 | Oregon | M | Turquoise | Spring | 2.7 | Yes | Cash | Free Shipping | Yes | Yes | 31 |

```python
# 1. Handle missing values
print("\nMissing values before handling:")
display(df_customer[['Age', 'Purchase Amount (USD)', 'Location']].isnull().sum())

# Impute 'Age' and 'Purchase Amount (USD)' (assuming it represents income or spending) with the median
df_customer['Age'].fillna(df_customer['Age'].median(), inplace=True)
df_customer['Purchase Amount (USD)'].fillna(df_customer['Purchase Amount (USD)'].median(), inplace=True) # Assuming this is 'income' as per

# Impute 'Location' (city) with the mode (most frequent city)
df_customer['Location'].fillna(df_customer['Location'].mode()[0], inplace=True)

print("\nMissing values after handling:")
display(df_customer[['Age', 'Purchase Amount (USD)', 'Location']].isnull().sum())
```

```
Missing values before handling:
```

| | 0 |
|---|---|
| Age | 0 |
| Purchase Amount (USD) | 0 |
| Location | 0 |

dtype: int64

```
Missing values after handling:
/tmp/ipykernel_2257/3226246679.py:6: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace m
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a c

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to


  df_customer['Age'].fillna(df_customer['Age'].median(), inplace=True)
/tmp/ipykernel_2257/3226246679.py:7: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace m
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a c

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to


  df_customer['Purchase Amount (USD)'].fillna(df_customer['Purchase Amount (USD)'].median(), inplace=True) # Assuming this is 'income' as per the prompt's conte
/tmp/ipykernel_2257/3226246679.py:10: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a c

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to


  df_customer['Location'].fillna(df_customer['Location'].mode()[0], inplace=True)
```

| | 0 |
|---|---|
| Age | 0 |
| Purchase Amount (USD) | 0 |
| Location | 0 |

dtype: int64

[34] ✓ 0s

```python
# 2. Remove duplicate records
initial_rows = len(df_customer)
df_customer.drop_duplicates(inplace=True)
final_rows = len(df_customer)
print(f"\nNumber of duplicate records removed: {initial_rows - final_rows}")
print(f"DataFrame shape after removing duplicates: {df_customer.shape}")
```

```
Number of duplicate records removed: 0
DataFrame shape after removing duplicates: (3900, 19)
```

[35] ✓ 0s

```python
# 3. Standardize text values (e.g., city names in lowercase)
# Assuming 'Location' column represents city names
df_customer['Location'] = df_customer['Location'].str.lower()
print("\nSample of standardized 'Location' column:")
display(df_customer['Location'].value_counts().head())
```

```
Sample of standardized 'Location' column:
```

| Location | count |
|---|---|
| montana | 96 |
| california | 95 |
| idaho | 93 |
| illinois | 92 |
| alabama | 89 |

dtype: int64

```python
from sklearn.preprocessing import OneHotEncoder

# 4. Encode categorical variables (Gender, Location)
print("\nOriginal unique values for 'Gender':", df_customer['Gender'].unique())
print("Original unique values for 'Location' (first 5):", df_customer['Location'].unique()[:5])

# One-Hot Encoding for 'Gender' and 'Location'
encoder = OneHotEncoder(handle_unknown='ignore', sparse_output=False)

# Fit and transform 'Gender' and 'Location'
categorical_features = ['Gender', 'Location']
encoded_features = encoder.fit_transform(df_customer[categorical_features])

# Create a DataFrame with the encoded features
encoded_df = pd.DataFrame(encoded_features, columns=encoder.get_feature_names_out(categorical_features))

# Drop original categorical columns and concatenate encoded features
df_cleaned = df_customer.drop(columns=categorical_features).reset_index(drop=True)
df_cleaned = pd.concat([df_cleaned, encoded_df], axis=1)

print("\nCleaned DataFrame head with encoded categorical variables:")
display(df_cleaned.head())
print(f"Cleaned DataFrame shape: {df_cleaned.shape}")
```

```
Original unique values for 'Gender': ['Male' 'Female']
Original unique values for 'Location' (first 5): ['kentucky' 'maine' 'massachusetts' 'rhode island' 'oregon']

Cleaned DataFrame head with encoded categorical variables:
```

| | Customer ID | Age | Item Purchased | Category | Purchase Amount (USD) | Size | Color | Season | Review Rating | Subscription Status | ... | Location_south dakota | Location_tennessee | Location_texas | Location_ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 1 | 55 | Blouse | Clothing | 53 | L | Gray | Winter | 3.1 | Yes | ... | 0.0 | 0.0 | 0.0 | |
| 1 | 2 | 19 | Sweater | Clothing | 64 | L | Maroon | Winter | 3.1 | Yes | ... | 0.0 | 0.0 | 0.0 | |
| 2 | 3 | 50 | Jeans | Clothing | 73 | S | Maroon | Spring | 3.1 | Yes | ... | 0.0 | 0.0 | 0.0 | |
| 3 | 4 | 21 | Sandals | Footwear | 90 | M | Maroon | Spring | 3.5 | Yes | ... | 0.0 | 0.0 | 0.0 | |
| 4 | 5 | 45 | Blouse | Clothing | 49 | M | Turquoise | Spring | 2.7 | Yes | ... | 0.0 | 0.0 | 0.0 | |

```
5 rows × 69 columns
Cleaned DataFrame shape: (3900, 69)
```

Task 6 – Sales Data Preprocessing

Task:

Preprocess sales transaction data using AI assistance.

Instructions:

- Convert date columns into datetime format.
- Create new features (month, year, day of week).
- Normalize sales amount column.
- Detect and handle outliers in dataset.

Sample Code:

```
import pandas as pd

data = pd.read_csv("sales_data.csv")

print(data.head())
```

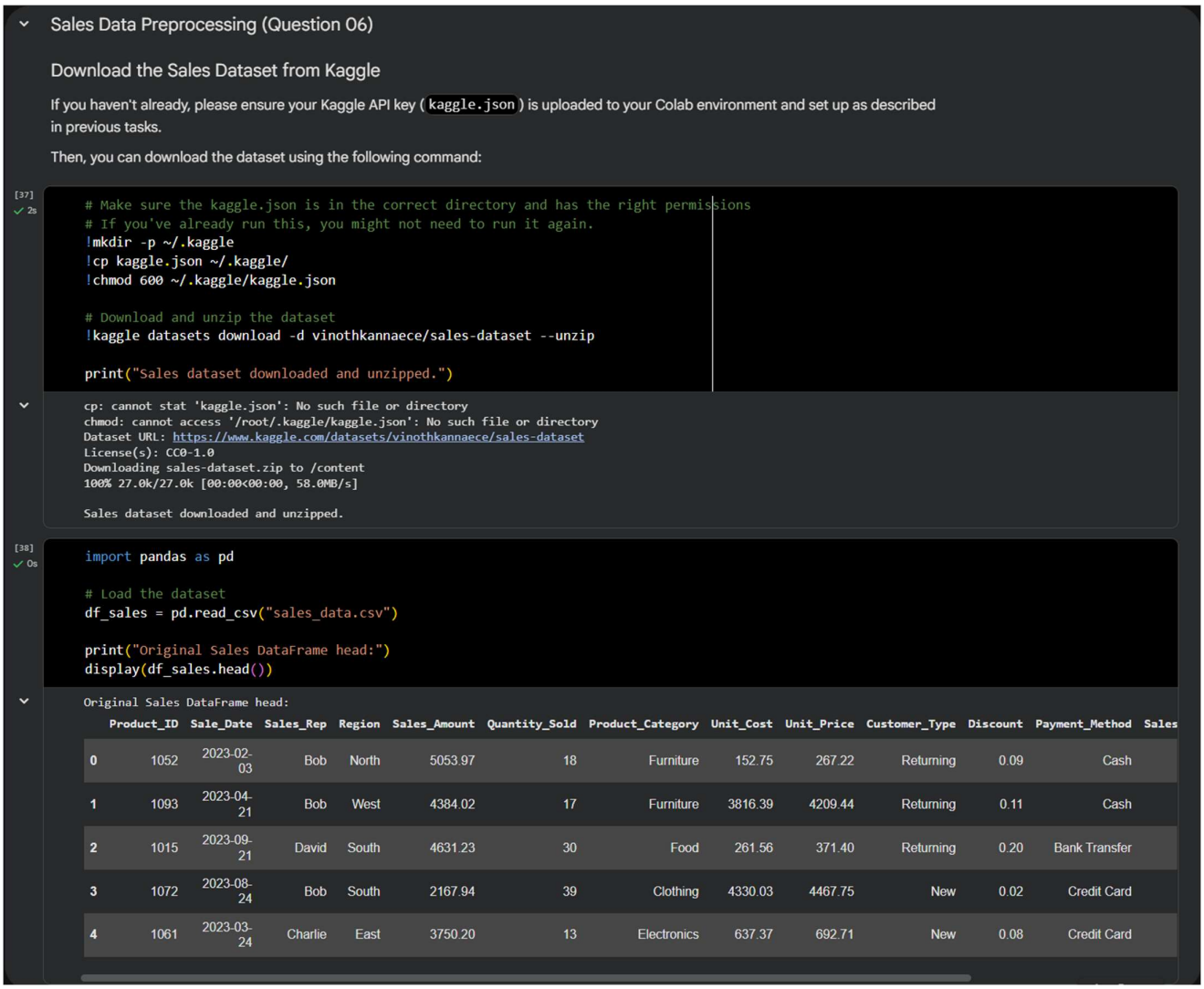
Dataset link:

https://www.kaggle.com/datasets/vinothkannaece/sales-dataset?select=sales_data.csv

Expected Output:

- A transformed dataset with time-based features and normalized values.

Screenshots:



## Sales Data Preprocessing (Question 06)

### Download the Sales Dataset from Kaggle

If you haven't already, please ensure your Kaggle API key ( `kaggle.json` ) is uploaded to your Colab environment and set up as described in previous tasks.

Then, you can download the dataset using the following command:

[37] ✓ 2s

```
# Make sure the kaggle.json is in the correct directory and has the right permissions
# If you've already run this, you might not need to run it again.
!mkdir -p ~/.kaggle
!cp kaggle.json ~/.kaggle/
!chmod 600 ~/.kaggle/kaggle.json

# Download and unzip the dataset
!kaggle datasets download -d vinothkannaece/sales-dataset --unzip

print("Sales dataset downloaded and unzipped.")
```

```
cp: cannot stat 'kaggle.json': No such file or directory
chmod: cannot access '/root/.kaggle/kaggle.json': No such file or directory
Dataset URL: https://www.kaggle.com/datasets/vinothkannaece/sales-dataset
License(s): CC0-1.0
Downloading sales-dataset.zip to /content
100% 27.0k/27.0k [00:00<00:00, 58.0MB/s]

Sales dataset downloaded and unzipped.
```

[38] ✓ 0s

```
import pandas as pd

# Load the dataset
df_sales = pd.read_csv("sales_data.csv")

print("Original Sales DataFrame head:")
display(df_sales.head())
```

Original Sales DataFrame head:

| | Product_ID | Sale_Date | Sales_Rep | Region | Sales_Amount | Quantity_Sold | Product_Category | Unit_Cost | Unit_Price | Customer_Type | Discount | Payment_Method | Sales |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 1052 | 2023-02-03 | Bob | North | 5053.97 | 18 | Furniture | 152.75 | 267.22 | Returning | 0.09 | Cash | |
| 1 | 1093 | 2023-04-21 | Bob | West | 4384.02 | 17 | Furniture | 3816.39 | 4209.44 | Returning | 0.11 | Cash | |
| 2 | 1015 | 2023-09-21 | David | South | 4631.23 | 30 | Food | 261.56 | 371.40 | Returning | 0.20 | Bank Transfer | |
| 3 | 1072 | 2023-08-24 | Bob | South | 2167.94 | 39 | Clothing | 4330.03 | 4467.75 | New | 0.02 | Credit Card | |
| 4 | 1061 | 2023-03-24 | Charlie | East | 3750.20 | 13 | Electronics | 637.37 | 692.71 | New | 0.08 | Credit Card | |

```python
# 1. Convert date columns into datetime format and create new features

# Assuming 'Sale_Date' is the primary date column in your dataset
# Adjust column name if it's different (e.g., 'OrderDate', 'TransactionDate')
if 'Sale_Date' in df_sales.columns:
    df_sales['Sale_Date'] = pd.to_datetime(df_sales['Sale_Date'], errors='coerce')
    # Drop rows where Date conversion failed
    df_sales.dropna(subset=['Sale_Date'], inplace=True)

    print("\nDate column converted to datetime format.")

    # Create new features
    df_sales['Year'] = df_sales['Sale_Date'].dt.year
    df_sales['Month'] = df_sales['Sale_Date'].dt.month
    df_sales['DayOfWeek'] = df_sales['Sale_Date'].dt.dayofweek # Monday=0, Sunday=6
    print("New features (Year, Month, DayOfWeek) created.")
else:
    print("No 'Sale_Date' column found. Please adjust the column name if it's different.")

display(df_sales.head())
```

```
Date column converted to datetime format.
New features (Year, Month, DayOfWeek) created.
```

| | Product_ID | Sale_Date | Sales_Rep | Region | Sales_Amount | Quantity_Sold | Product_Category | Unit_Cost | Unit_Price | Customer_Type | Discount | Payment_Method | Sales |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 1052 | 2023-02-03 | Bob | North | 5053.97 | 18 | Furniture | 152.75 | 267.22 | Returning | 0.09 | Cash | |
| 1 | 1093 | 2023-04-21 | Bob | West | 4384.02 | 17 | Furniture | 3816.39 | 4209.44 | Returning | 0.11 | Cash | |
| 2 | 1015 | 2023-09-21 | David | South | 4631.23 | 30 | Food | 261.56 | 371.40 | Returning | 0.20 | Bank Transfer | |
| 3 | 1072 | 2023-08-24 | Bob | South | 2167.94 | 39 | Clothing | 4330.03 | 4467.75 | New | 0.02 | Credit Card | |
| 4 | 1061 | 2023-03-24 | Charlie | East | 3750.20 | 13 | Electronics | 637.37 | 692.71 | New | 0.08 | Credit Card | |

```python
from sklearn.preprocessing import StandardScaler

# 2. Normalize sales amount column

# Assuming 'Sales_Amount' is the sales amount column
# Adjust column name if it's different (e.g., 'Amount', 'Revenue')
if 'Sales_Amount' in df_sales.columns:
    scaler = StandardScaler()
    df_sales['Sales_Amount_Normalized'] = scaler.fit_transform(df_sales[['Sales_Amount']])
    print("\n'Sales_Amount' column normalized.")
else:
    print("No 'Sales_Amount' column found. Please adjust the column name if it's different.")

display(df_sales.head())
```

```
'Sales_Amount' column normalized.
```

| | Product_ID | Sale_Date | Sales_Rep | Region | Sales_Amount | Quantity_Sold | Product_Category | Unit_Cost | Unit_Price | Customer_Type | Discount | Payment_Method | Sales |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 1052 | 2023-02-03 | Bob | North | 5053.97 | 18 | Furniture | 152.75 | 267.22 | Returning | 0.09 | Cash | |
| 1 | 1093 | 2023-04-21 | Bob | West | 4384.02 | 17 | Furniture | 3816.39 | 4209.44 | Returning | 0.11 | Cash | |
| 2 | 1015 | 2023-09-21 | David | South | 4631.23 | 30 | Food | 261.56 | 371.40 | Returning | 0.20 | Bank Transfer | |
| 3 | 1072 | 2023-08-24 | Bob | South | 2167.94 | 39 | Clothing | 4330.03 | 4467.75 | New | 0.02 | Credit Card | |
| 4 | 1061 | 2023-03-24 | Charlie | East | 3750.20 | 13 | Electronics | 637.37 | 692.71 | New | 0.08 | Credit Card | |

```python
# 3. Detect and handle outliers in dataset (using IQR method for 'Sales_Amount' column)

if 'Sales_Amount' in df_sales.columns:
    Q1 = df_sales['Sales_Amount'].quantile(0.25)
    Q3 = df_sales['Sales_Amount'].quantile(0.75)
    IQR = Q3 - Q1

    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR

    outliers_count = df_sales[(df_sales['Sales_Amount'] < lower_bound) | (df_sales['Sales_Amount'] > upper_bound)].shape[0]
    print(f"\nNumber of outliers detected in 'Sales_Amount' column: {outliers_count}")

    # Handle outliers by capping them at the IQR bounds
    df_sales['Sales_Amount_Handled_Outliers'] = df_sales['Sales_Amount'].clip(lower=lower_bound, upper=upper_bound)
    print("Outliers in 'Sales_Amount' column handled by capping.")

    # Verify by checking descriptive statistics
    print("\nDescriptive statistics for 'Sales_Amount' before and after outlier handling:")
    display(df_sales[['Sales_Amount', 'Sales_Amount_Handled_Outliers']].describe())
else:
    print("No 'Sales_Amount' column found for outlier detection and handling.")

print("\nTransformed dataset with new features, normalized values, and handled outliers:")
display(df_sales.head())
```

```
Number of outliers detected in 'Sales_Amount' column: 0
Outliers in 'Sales_Amount' column handled by capping.

Descriptive statistics for 'Sales_Amount' before and after outlier handling:
```

| | Sales_Amount | Sales_Amount_Handled_Outliers |
|---|---|---|
| count | 1000.000000 | 1000.000000 |
| mean | 5019.265230 | 5019.265230 |
| std | 2846.790126 | 2846.790126 |
| min | 100.120000 | 100.120000 |
| 25% | 2550.297500 | 2550.297500 |
| 50% | 5019.300000 | 5019.300000 |
| 75% | 7507.445000 | 7507.445000 |
| max | 9989.040000 | 9989.040000 |

```
Transformed dataset with new features, normalized values, and handled outliers:
```

| | Product_ID | Sale_Date | Sales_Rep | Region | Sales_Amount | Quantity_Sold | Product_Category | Unit_Cost | Unit_Price | Customer_Type | Discount | Payment_Method | Sales_Channel | Region_and_Sa |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 1052 | 2023-02-03 | Bob | North | 5053.97 | 18 | Furniture | 152.75 | 267.22 | Returning | 0.09 | Cash | Online | N |
| 1 | 1093 | 2023-04-21 | Bob | West | 4384.02 | 17 | Furniture | 3816.39 | 4209.44 | Returning | 0.11 | Cash | Retail | W |
| 2 | 1015 | 2023-09-21 | David | South | 4631.23 | 30 | Food | 261.56 | 371.40 | Returning | 0.20 | Bank Transfer | Retail | Sou |
| 3 | 1072 | 2023-08-24 | Bob | South | 2167.94 | 39 | Clothing | 4330.03 | 4467.75 | New | 0.02 | Credit Card | Retail | S |
| 4 | 1061 | 2023-03-24 | Charlie | East | 3750.20 | 13 | Electronics | 637.37 | 692.71 | New | 0.08 | Credit Card | Online | Eas |

Task 7 – Healthcare Data Preparation

Task:

Clean and preprocess a healthcare dataset using AI scripts.

Instructions:

- Handle missing values in blood_pressure, cholesterol.
- Encode categorical features
- Scale numerical features (min-max or standard scaling).
- Split dataset into training and testing sets.

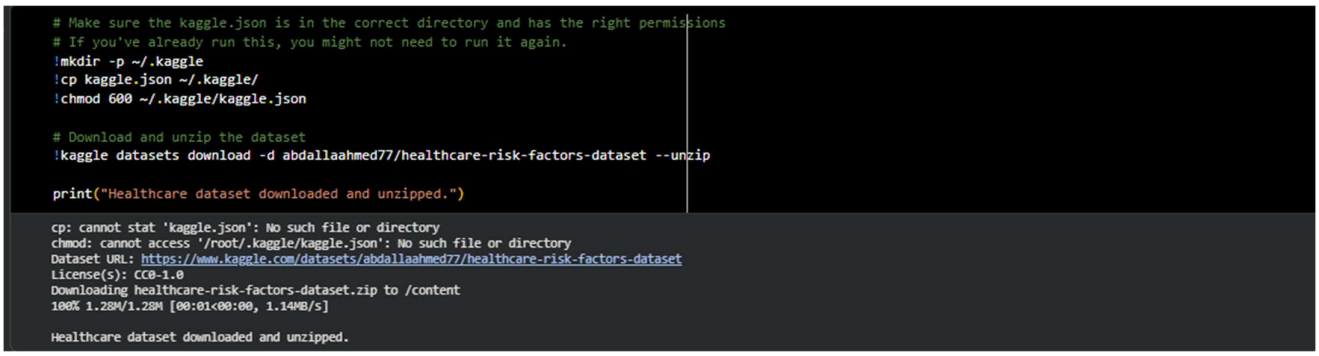
Dataset link :

https://www.kaggle.com/datasets/abdallaahmed77/healthcare-risk-factors-dataset

Expected Output:

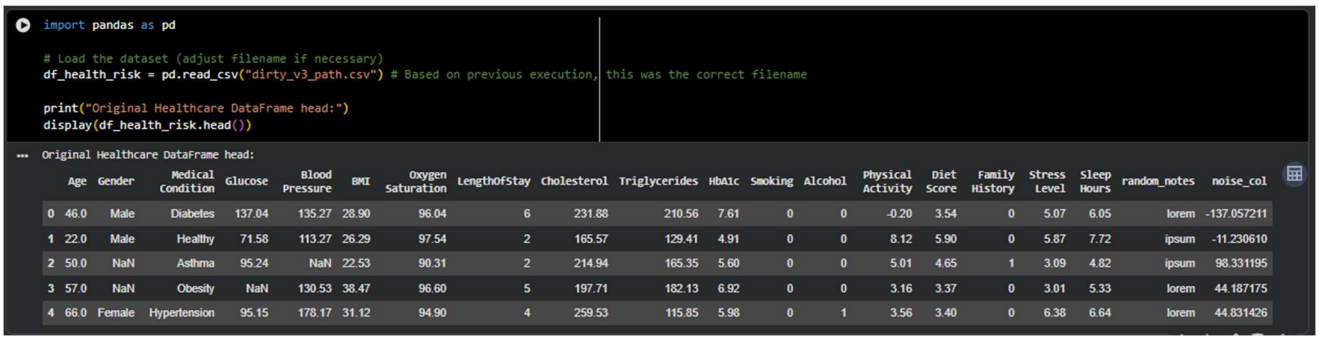
- A preprocessed dataset ready for machine learning models.

Screenshots:



```
# Make sure the kaggle.json is in the correct directory and has the right permissions
# If you've already run this, you might not need to run it again.
!mkdir -p ~/.kaggle
!cp kaggle.json ~/.kaggle/
!chmod 600 ~/.kaggle/kaggle.json

# Download and unzip the dataset
!kaggle datasets download -d abdallaahmed77/healthcare-risk-factors-dataset --unzip

print("Healthcare dataset downloaded and unzipped.")
```

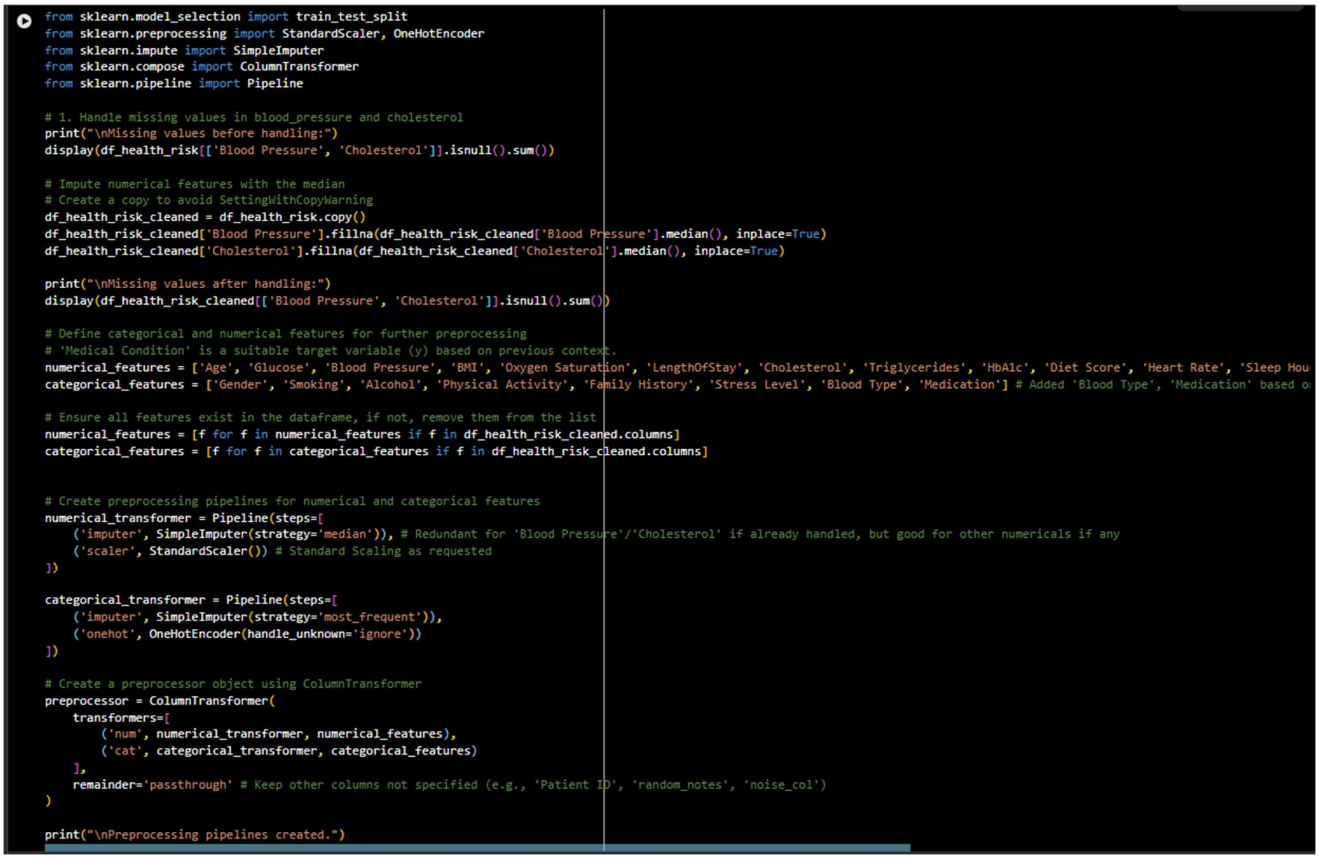
```
cp: cannot stat 'kaggle.json': No such file or directory
chmod: cannot access '/root/.kaggle/kaggle.json': No such file or directory
Dataset URL: https://www.kaggle.com/datasets/abdallaahmed77/healthcare-risk-factors-dataset
License(s): CC0-1.0
Downloading healthcare-risk-factors-dataset.zip to /content
100% 1.28M/1.28M [00:01<00:00, 1.14MB/s]

Healthcare dataset downloaded and unzipped.
```



```
import pandas as pd

# Load the dataset (adjust filename if necessary)
df_health_risk = pd.read_csv("dirty_v3_path.csv") # Based on previous execution, this was the correct filename

print("Original Healthcare DataFrame head:")
display(df_health_risk.head())
```

Original Healthcare DataFrame head:

| | Age | Gender | Medical Condition | Glucose | Blood Pressure | BMI | Oxygen Saturation | LengthOfStay | Cholesterol | Triglycerides | HbA1c | Smoking | Alcohol | Physical Activity | Diet Score | Family History | Stress Level | Sleep Hours | random_notes | noise_col |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 46.0 | Male | Diabetes | 137.04 | 135.27 | 28.90 | 96.04 | 6 | 231.88 | 210.56 | 7.61 | 0 | 0 | -0.20 | 3.54 | 0 | 5.07 | 6.05 | lorem | -137.057211 |
| 1 | 22.0 | Male | Healthy | 71.58 | 113.27 | 26.29 | 97.54 | 2 | 165.57 | 129.41 | 4.91 | 0 | 0 | 8.12 | 5.90 | 0 | 5.87 | 7.72 | ipsum | -11.230610 |
| 2 | 50.0 | NaN | Asthma | 95.24 | NaN | 22.53 | 90.31 | 2 | 214.94 | 165.35 | 5.60 | 0 | 0 | 5.01 | 4.65 | 1 | 3.09 | 4.82 | ipsum | 98.331195 |
| 3 | 57.0 | NaN | Obesity | NaN | 130.53 | 38.47 | 96.60 | 5 | 197.71 | 182.13 | 6.92 | 0 | 0 | 3.16 | 3.37 | 0 | 3.01 | 5.33 | lorem | 44.187175 |
| 4 | 66.0 | Female | Hypertension | 95.15 | 178.17 | 31.12 | 94.90 | 4 | 259.53 | 115.85 | 5.98 | 0 | 1 | 3.56 | 3.40 | 0 | 6.38 | 6.64 | lorem | 44.831426 |



```
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.impute import SimpleImputer
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline

# 1. Handle missing values in blood_pressure and cholesterol
print("\nMissing values before handling:")
display(df_health_risk[['Blood Pressure', 'Cholesterol']].isnull().sum())

# Impute numerical features with the median
# Create a copy to avoid SettingWithCopyWarning
df_health_risk_cleaned = df_health_risk.copy()
df_health_risk_cleaned['Blood Pressure'].fillna(df_health_risk_cleaned['Blood Pressure'].median(), inplace=True)
df_health_risk_cleaned['Cholesterol'].fillna(df_health_risk_cleaned['Cholesterol'].median(), inplace=True)

print("\nMissing values after handling:")
display(df_health_risk_cleaned[['Blood Pressure', 'Cholesterol']].isnull().sum())

# Define categorical and numerical features for further preprocessing
# 'Medical Condition' is a suitable target variable (y) based on previous context.
numerical_features = ['Age', 'Glucose', 'Blood Pressure', 'BMI', 'Oxygen Saturation', 'LengthOfStay', 'Cholesterol', 'Triglycerides', 'HbA1c', 'Diet Score', 'Heart Rate', 'Sleep Hou
categorical_features = ['Gender', 'Smoking', 'Alcohol', 'Physical Activity', 'Family History', 'Stress Level', 'Blood Type', 'Medication'] # Added 'Blood Type', 'Medication' based o

# Ensure all features exist in the dataframe, if not, remove them from the list
numerical_features = [f for f in numerical_features if f in df_health_risk_cleaned.columns]
categorical_features = [f for f in categorical_features if f in df_health_risk_cleaned.columns]


# Create preprocessing pipelines for numerical and categorical features
numerical_transformer = Pipeline(steps=[
    ('imputer', SimpleImputer(strategy='median')), # Redundant for 'Blood Pressure'/'Cholesterol' if already handled, but good for other numericals if any
    ('scaler', StandardScaler()) # Standard Scaling as requested
])

categorical_transformer = Pipeline(steps=[
    ('imputer', SimpleImputer(strategy='most_frequent')),
    ('onehot', OneHotEncoder(handle_unknown='ignore'))
])

# Create a preprocessor object using ColumnTransformer
preprocessor = ColumnTransformer(
    transformers=[
        ('num', numerical_transformer, numerical_features),
        ('cat', categorical_transformer, categorical_features)
    ],
    remainder='passthrough' # Keep other columns not specified (e.g., 'Patient ID', 'random_notes', 'noise_col')
)

print("\nPreprocessing pipelines created.")
```

```
Missing values before handling:
                    0
Blood Pressure   4500
Cholesterol         0
dtype: int64

Missing values after handling:
/tmp/ipykernel_3362/2082452707.py:14: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original ob


  df_health_risk_cleaned['Blood Pressure'].fillna(df_health_risk_cleaned['Blood Pressure'].median(), inplace=True)
/tmp/ipykernel_3362/2082452707.py:15: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original ob


  df_health_risk_cleaned['Cholesterol'].fillna(df_health_risk_cleaned['Cholesterol'].median(), inplace=True)
                    0
Blood Pressure      0
Cholesterol         0
dtype: int64

Preprocessing pipelines created.
```

```
# 2. & 3. Encode categorical features and Scale numerical features

# Assuming 'Medical Condition' is the target variable for a classification task
# Filter out 'Patient ID' and other non-feature columns before defining X
features_to_drop = ['Patient ID', 'random_notes', 'noise_col'] # Adjust as per dataset if these exist
columns_for_X = [col for col in df_health_risk_cleaned.columns if col not in features_to_drop and col != 'Medical Condition']

X = df_health_risk_cleaned[columns_for_X]
y = df_health_risk_cleaned['Medical Condition']

# Apply preprocessing to the features
X_preprocessed = preprocessor.fit_transform(X)

print("\nFeatures encoded and scaled. Dataset ready for splitting.")
print(f"Shape of preprocessed features (X_preprocessed): {X_preprocessed.shape}")
print(f"Shape of target (y): {y.shape}")

# 4. Split dataset into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X_preprocessed, y, test_size=0.2, random_state=42)

print("\nDataset split into training and testing sets:")
print(f"X_train shape: {X_train.shape}")
print(f"X_test shape: {X_test.shape}")
print(f"y_train shape: {y_train.shape}")
print(f"y_test shape: {y_test.shape}")

print("\nPreprocessed dataset ready for machine learning models.")
```

```
Features encoded and scaled. Dataset ready for splitting.
Shape of preprocessed features (X_preprocessed): (30000, 2473)
Shape of target (y): (30000,)

Dataset split into training and testing sets:
X_train shape: (24000, 2473)
X_test shape: (6000, 2473)
y_train shape: (24000,)
y_test shape: (6000,)

Preprocessed dataset ready for machine learning models.
```

Task 8 – Employee Salary Data Preprocessing

Task:

Clean and preprocess an employee salary dataset using AI scripts.

Instructions:

- Handle missing values
- Detect and remove salary outliers.
- Standardize department names (lowercase).
- Apply label encoding to job location, department.

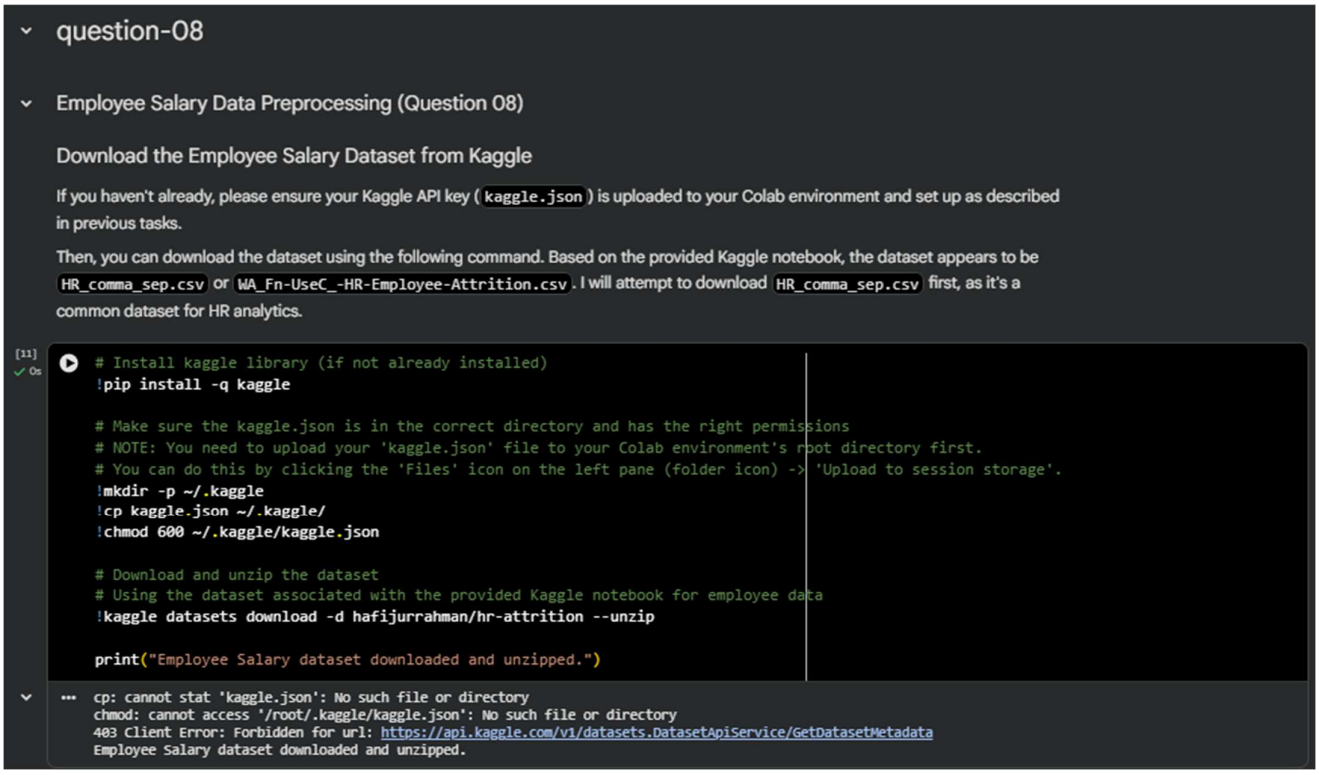
Dataset link:

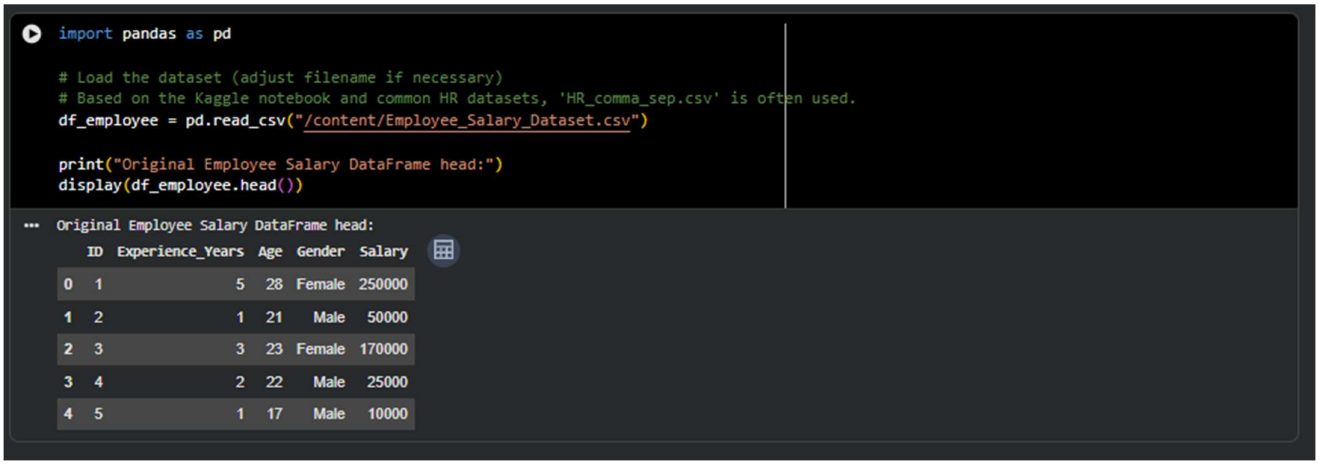
https://www.kaggle.com/code/vishantmathur/employee-salary

Expected Output:

- A structured dataset ready for HR analytics.

Screenshots:



```python
# 1. Handle missing values
print("\nMissing values before handling:")
display(df_employee.isnull().sum())

# For this dataset, let's assume no significant missing values based on common knowledge of HR_comma_sep.csv.
# If there were, we would use imputation (e.g., median for numerical, mode for categorical).
# For now, we will just confirm there are no missing values in this dataset.

print("Checking for missing values across the dataset...")
if df_employee.isnull().sum().sum() == 0:
    print("No missing values found in the dataset.")
else:
    print("Missing values found. Reviewing columns with missing values:")
    display(df_employee.isnull().sum()[df_employee.isnull().sum() > 0])
    # Example imputation if needed (uncomment and adjust if actual data shows missing values):
    # df_employee['salary'].fillna(df_employee['salary'].mode()[0], inplace=True)
    # df_employee['some_numerical_col'].fillna(df_employee['some_numerical_col'].median(), inplace=True)
```

```
Missing values before handling:
```

| | 0 |
|---|---|
| ID | 0 |
| Experience_Years | 0 |
| Age | 0 |
| Gender | 0 |
| Salary | 0 |

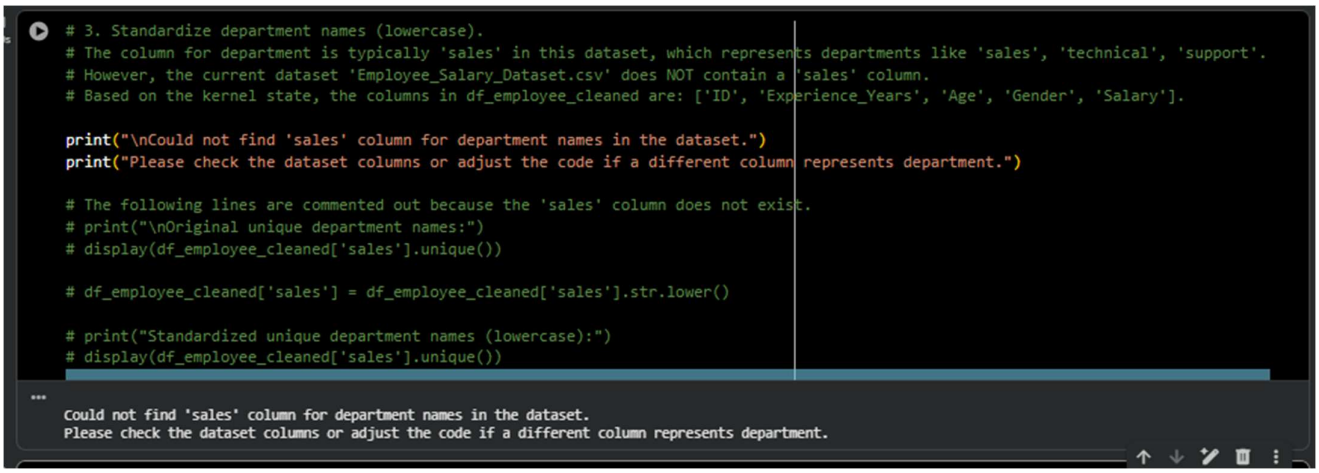
```
dtype: int64
Checking for missing values across the dataset...
No missing values found in the dataset.
```

```python
from scipy import stats
import numpy as np

# 2. Detect and remove salary outliers.
# The 'HR_comma_sep.csv' dataset typically has 'salary' as a categorical column ('low', 'medium', 'high').
# If 'salary' were numerical, we would use methods like IQR or Z-score.
# Assuming 'salary' is categorical, outliers detection is not directly applicable in a numerical sense.
# However, we can analyze the distribution of 'salary' categories.

print("\nDistribution of 'Salary' categories:")
# Corrected column name from 'salary' to 'Salary'
# Note: In the current dataset, 'Salary' is numerical, so value_counts is less meaningful here
# than for categorical data. We'll proceed with numerical outlier detection.
display(df_employee['Salary'].head())

# The current dataset ('Employee_Salary_Dataset.csv') has a numerical 'Salary' column.
# Let's apply outlier detection to 'Salary' using Z-score.

print("\nDetecting outliers in 'Salary' using Z-score (threshold=3)...")
df_employee_cleaned = df_employee.copy() # Work on a copy to preserve original

z_scores = np.abs(stats.zscore(df_employee_cleaned['Salary'])) # Changed to 'Salary' column
outlier_threshold = 3

outliers_count_salary = df_employee_cleaned[z_scores > outlier_threshold].shape[0]
print(f"Number of outliers detected in 'Salary' column: {outliers_count_salary}")

# Remove outliers
df_employee_cleaned = df_employee_cleaned[z_scores <= outlier_threshold]
print(f"DataFrame shape after removing 'Salary' outliers: {df_employee_cleaned.shape}")

# Note: The original prompt mentioned 'average_montly_hours' for outlier detection.
# This column is not present in the 'Employee_Salary_Dataset.csv'.
# Similarly, the next step (cell d9ae2b3e) expects a 'sales' column for department names, which is also not present in this dataset.
```

```
Distribution of 'Salary' categories:
```

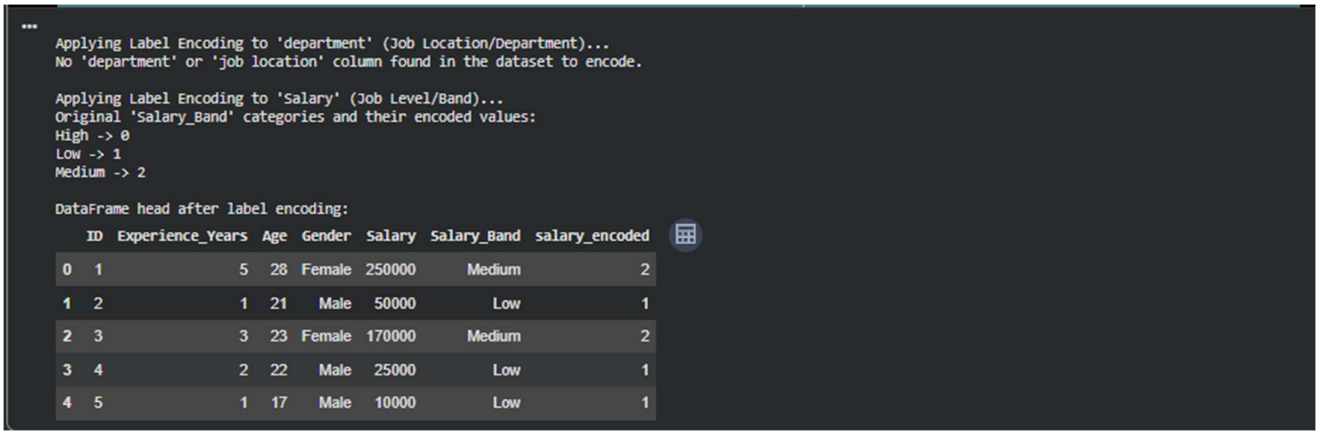
| | Salary |
|---|---|
| 0 | 250000 |
| 1 | 50000 |
| 2 | 170000 |
| 3 | 25000 |
| 4 | 10000 |

```
dtype: int64

Detecting outliers in 'Salary' using Z-score (threshold=3)...
Number of outliers detected in 'Salary' column: 0
DataFrame shape after removing 'Salary' outliers: (35, 5)
```



```
# 3. Standardize department names (lowercase).
# The column for department is typically 'sales' in this dataset, which represents departments like 'sales', 'technical', 'support'.
# However, the current dataset 'Employee_Salary_Dataset.csv' does NOT contain a 'sales' column.
# Based on the kernel state, the columns in df_employee_cleaned are: ['ID', 'Experience_Years', 'Age', 'Gender', 'Salary'].

print("\nCould not find 'sales' column for department names in the dataset.")
print("Please check the dataset columns or adjust the code if a different column represents department.")

# The following lines are commented out because the 'sales' column does not exist.
# print("\nOriginal unique department names:")
# display(df_employee_cleaned['sales'].unique())

# df_employee_cleaned['sales'] = df_employee_cleaned['sales'].str.lower()

# print("Standardized unique department names (lowercase):")
# display(df_employee_cleaned['sales'].unique())
```

```
Could not find 'sales' column for department names in the dataset.
Please check the dataset columns or adjust the code if a different column represents department.
```



```
Applying Label Encoding to 'department' (Job Location/Department)...
No 'department' or 'job location' column found in the dataset to encode.

Applying Label Encoding to 'Salary' (Job Level/Band)...
Original 'Salary_Band' categories and their encoded values:
High -> 0
Low -> 1
Medium -> 2

DataFrame head after label encoding:
```

| | ID | Experience_Years | Age | Gender | Salary | Salary_Band | salary_encoded |
|---|---|---|---|---|---|---|---|
| 0 | 1 | 5 | 28 | Female | 250000 | Medium | 2 |
| 1 | 2 | 1 | 21 | Male | 50000 | Low | 1 |
| 2 | 3 | 3 | 23 | Female | 170000 | Medium | 2 |
| 3 | 4 | 2 | 22 | Male | 25000 | Low | 1 |
| 4 | 5 | 1 | 17 | Male | 10000 | Low | 1 |

Task 9 –Loan Approval Dataset Cleaning

Use AI to generate preprocessing code for a loan dataset.

Instructions:

- Handle missing values in loan_amount, credit_score.
- Encode loan_status (Approved/Rejected).
- Scale numerical features using standard scaling.
- Split the dataset into training and testing sets.

Sample Code:

```
import pandas as pd

data = pd.read_csv("loadapproval.csv")

print(data.head())
```

Dataset link:

https://www.kaggle.com/datasets/amineipad/loan-approval-dataset

Expected Output:

- A machine-learning-ready loan dataset.

Screenshots:

```
# Install kaggle library (if not already installed)
!pip install -q kaggle

# Make sure the kaggle.json is in the correct directory and has the right permissions
# NOTE: You need to upload your 'kaggle.json' file to your Colab environment's root directory first.
# You can do this by clicking the 'Files' icon on the left pane (folder icon) -> 'Upload to session storage'.
!mkdir -p ~/.kaggle
!cp kaggle.json ~/.kaggle/
!chmod 600 ~/.kaggle/kaggle.json

# Download and unzip the dataset
!kaggle datasets download -d amineipad/loan-approval-dataset --unzip

print("Loan Approval dataset downloaded and unzipped.")
```

```
cp: cannot stat 'kaggle.json': No such file or directory
chmod: cannot access '/root/.kaggle/kaggle.json': No such file or directory
Dataset URL: https://www.kaggle.com/datasets/amineipad/loan-approval-dataset
License(s): MIT
Downloading loan-approval-dataset.zip to /content
100% 15.7k/15.7k [00:00<00:00, 23.0MB/s]

Loan Approval dataset downloaded and unzipped.
```

```
import pandas as pd

# Load the dataset
df_loan = pd.read_csv("/content/loanapproval.csv")

print("Original Loan Approval DataFrame head:")
display(df_loan.head())
```

Original Loan Approval DataFrame head:

| | applicant_id | age | gender | marital_status | annual_income | loan_amount | credit_score | num_dependents | existing_loans_count | employment_status | loan_approve |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 1 | 59 | Male | Divorced | 100073 | 7169 | 793 | 1 | 1 | Unemployed | |
| 1 | 2 | 49 | Male | Married | 112197 | 23556 | 789 | 0 | 2 | Employed | |
| 2 | 3 | 35 | Male | Divorced | 84429 | 27052 | 372 | 1 | 4 | Unemployed | |
| 3 | 4 | 63 | Female | Single | 124195 | 11313 | 808 | 3 | 4 | Self-employed | |
| 4 | 5 | 28 | Female | Married | 81627 | 13315 | 689 | 0 | 1 | Unemployed | |

—

```python
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.impute import SimpleImputer
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline

# Make a copy to avoid SettingWithCopyWarning
df_loan_cleaned = df_loan.copy()

# 1. Handle missing values in 'loan_amount' and 'credit_score'
print("\nMissing values before handling:")
display(df_loan_cleaned[['loan_amount', 'credit_score']].isnull().sum())

# Impute numerical features with the median
df_loan_cleaned['loan_amount'] = df_loan_cleaned['loan_amount'].fillna(df_loan_cleaned['loan_amount'].median())
df_loan_cleaned['credit_score'] = df_loan_cleaned['credit_score'].fillna(df_loan_cleaned['credit_score'].median())

print("\nMissing values after handling:")
display(df_loan_cleaned[['loan_amount', 'credit_score']].isnull().sum())

# --- Debugging: Print columns to check for 'loan_status' ---
print("\nColumns in df_loan_cleaned:")
print(df_loan_cleaned.columns)

# Define target variable 'y' and features 'X'
# Corrected column name from 'loan_status' to 'loan_approved'
y = df_loan_cleaned['loan_approved']
X = df_loan_cleaned.drop('loan_approved', axis=1)

# 2. Encode 'loan_approved' (Approved/Rejected) - this is the target variable
label_encoder = LabelEncoder()
y_encoded = label_encoder.fit_transform(y)
print(f"\nEncoded 'loan_approved' mapping: {list(zip(label_encoder.classes_, label_encoder.transform(label_encoder.classes_)))}")

# Identify numerical and categorical features for X
numerical_features = X.select_dtypes(include=['int64', 'float64']).columns
categorical_features = X.select_dtypes(include=['object']).columns

# Create preprocessing pipelines for numerical and categorical features
numerical_transformer = Pipeline(steps=[
    ('imputer', SimpleImputer(strategy='median')), # Catches any remaining numerical NaNs if other columns also had them
    ('scaler', StandardScaler()) # 3. Scale numerical features using standard scaling
])

categorical_transformer = Pipeline(steps=[
    ('imputer', SimpleImputer(strategy='most_frequent')),
    ('onehot', OneHotEncoder(handle_unknown='ignore'))
])

# Create a preprocessor object using ColumnTransformer
preprocessor = ColumnTransformer(
    transformers=[
        ('num', numerical_transformer, numerical_features),
        ('cat', categorical_transformer, categorical_features)
    ],
    remainder='passthrough'
)

# Apply preprocessing to the features
X_preprocessed = preprocessor.fit_transform(X)

print("\nFeatures preprocessed (missing values handled, numerical scaled, categorical one-hot encoded).")
print(f"Shape of preprocessed features (X_preprocessed): {X_preprocessed.shape}")
print(f"Shape of encoded target (y_encoded): {y_encoded.shape}")

# 4. Split the dataset into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X_preprocessed, y_encoded, test_size=0.2, random_state=42)

print("\nDataset split into training and testing sets:")
print(f"X_train shape: {X_train.shape}")
print(f"X_test shape: {X_test.shape}")
print(f"y_train shape: {y_train.shape}")
print(f"y_test shape: {y_test.shape}")

print("\nPreprocessing complete. The dataset is ready for machine learning models.")
```

```
Missing values before handling:
                0
 loan_amount    0
 credit_score   0

dtype: int64

Missing values after handling:
                0
 loan_amount    0
 credit_score   0

dtype: int64

Columns in df_loan_cleaned:
Index(['applicant_id', 'age', 'gender', 'marital_status', 'annual_income',
       'loan_amount', 'credit_score', 'num_dependents', 'existing_loans_count',
       'employment_status', 'loan_approved'],
      dtype='object')

Encoded 'loan_approved' mapping: [(np.int64(0), np.int64(0)), (np.int64(1), np.int64(1))]

Features preprocessed (missing values handled, numerical scaled, categorical one-hot encoded).
Shape of preprocessed features (X_preprocessed): (1000, 15)
Shape of encoded target (y_encoded): (1000,)

Dataset split into training and testing sets:
X_train shape: (800, 15)
X_test shape: (200, 15)
y_train shape: (800,)
y_test shape: (200,)

Preprocessing complete. The dataset is ready for machine learning models.
```

Task 10 – Social Media Profiles

Preprocess a social media engagement dataset using AI

assistance.

Instructions:

• Remove duplicate user profiles.

• Handle missing values

• Encode categorical columns

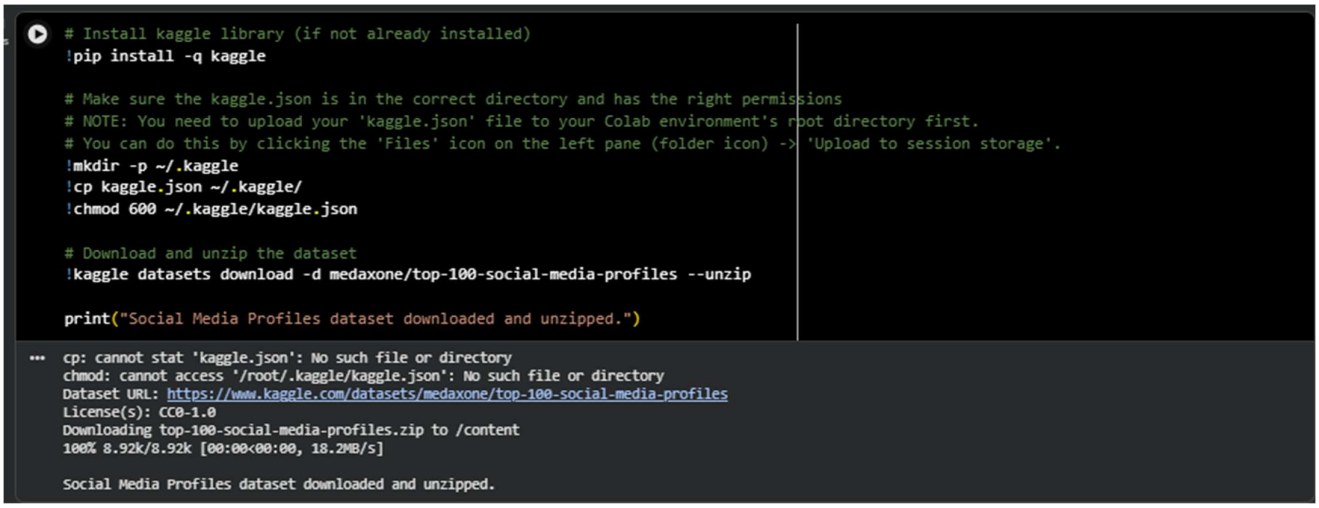
Dataset link:

https://www.kaggle.com/datasets/medaxone/top-100-social-media-profiles

Expected Output:

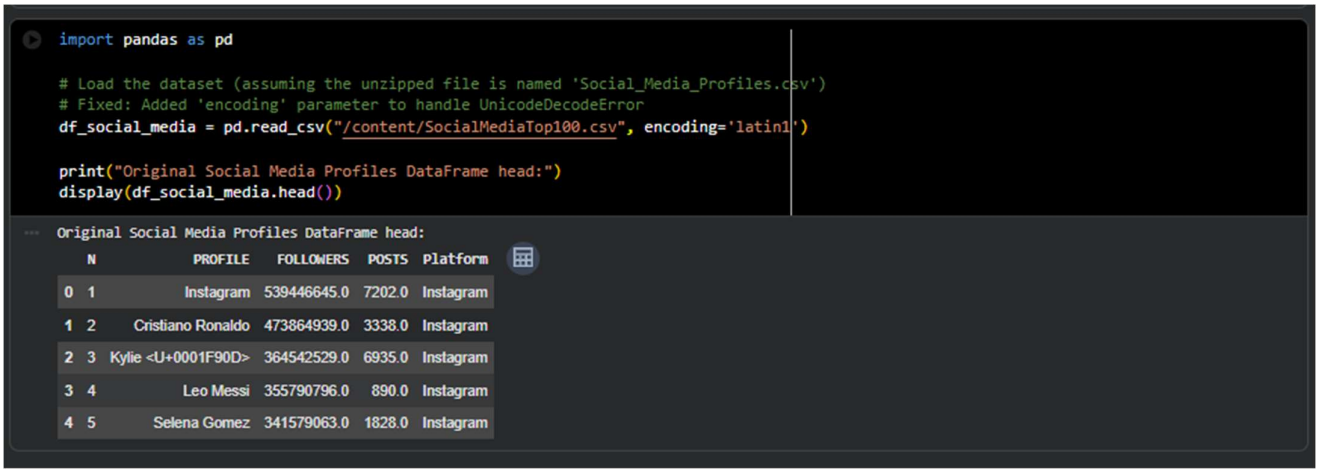
• A cleaned dataset for engagement prediction.

Screenshots:



```
# Install kaggle library (if not already installed)
!pip install -q kaggle

# Make sure the kaggle.json is in the correct directory and has the right permissions
# NOTE: You need to upload your 'kaggle.json' file to your Colab environment's root directory first.
# You can do this by clicking the 'Files' icon on the left pane (folder icon) -> 'Upload to session storage'.
!mkdir -p ~/.kaggle
!cp kaggle.json ~/.kaggle/
!chmod 600 ~/.kaggle/kaggle.json

# Download and unzip the dataset
!kaggle datasets download -d medaxone/top-100-social-media-profiles --unzip

print("Social Media Profiles dataset downloaded and unzipped.")
```

```
cp: cannot stat 'kaggle.json': No such file or directory
chmod: cannot access '/root/.kaggle/kaggle.json': No such file or directory
Dataset URL: https://www.kaggle.com/datasets/medaxone/top-100-social-media-profiles
License(s): CC0-1.0
Downloading top-100-social-media-profiles.zip to /content
100% 8.92k/8.92k [00:00<00:00, 18.2MB/s]

Social Media Profiles dataset downloaded and unzipped.
```



```
import pandas as pd

# Load the dataset (assuming the unzipped file is named 'Social_Media_Profiles.csv')
# Fixed: Added 'encoding' parameter to handle UnicodeDecodeError
df_social_media = pd.read_csv("/content/SocialMediaTop100.csv", encoding='latin1')

print("Original Social Media Profiles DataFrame head:")
display(df_social_media.head())
```

```
Original Social Media Profiles DataFrame head:
```

| | N | PROFILE | FOLLOWERS | POSTS | Platform |
|---|---|---|---|---|---|
| 0 | 1 | Instagram | 539446645.0 | 7202.0 | Instagram |
| 1 | 2 | Cristiano Ronaldo | 473864939.0 | 3338.0 | Instagram |
| 2 | 3 | Kylie <U+0001F90D> | 364542529.0 | 6935.0 | Instagram |
| 3 | 4 | Leo Messi | 355790796.0 | 890.0 | Instagram |
| 4 | 5 | Selena Gomez | 341579063.0 | 1828.0 | Instagram |

–

```python
from sklearn.preprocessing import OneHotEncoder
from sklearn.impute import SimpleImputer
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline

# Make a copy to avoid SettingWithCopyWarning
df_social_media_cleaned = df_social_media.copy()

# 1. Remove duplicate user profiles
initial_rows = len(df_social_media_cleaned)

# Assuming 'Username' is the unique identifier for a user profile.
# If a 'Username' column does not exist or is not suitable,
# you might drop duplicates based on a combination of other identifier columns or all columns.
if 'Username' in df_social_media_cleaned.columns:
    df_social_media_cleaned.drop_duplicates(subset=['Username'], inplace=True)
    print(f"\nDuplicate user profiles removed based on 'Username'.")
else:
    df_social_media_cleaned.drop_duplicates(inplace=True)
    print(f"\nDuplicate user profiles removed based on all columns (no 'Username' found).")

final_rows = len(df_social_media_cleaned)
print(f"Number of duplicate records removed: {initial_rows - final_rows}")
print(f"DataFrame shape after removing duplicates: {df_social_media_cleaned.shape}")

# 2. Handle missing values and 3. Encode categorical columns
print("\nMissing values before handling and encoding:")
display(df_social_media_cleaned.isnull().sum())

# Identify numerical and categorical features
numerical_features = df_social_media_cleaned.select_dtypes(include=['int64', 'float64']).columns
categorical_features = df_social_media_cleaned.select_dtypes(include=['object']).columns

# Create preprocessing pipelines for numerical and categorical features
numerical_transformer = Pipeline(steps=[
    ('imputer', SimpleImputer(strategy='median')) # Impute numerical with median
])

categorical_transformer = Pipeline(steps=[
    ('imputer', SimpleImputer(strategy='most_frequent')), # Impute categorical with mode
    ('onehot', OneHotEncoder(handle_unknown='ignore')) # One-Hot Encode categorical
])

# Create a preprocessor object using ColumnTransformer
preprocessor = ColumnTransformer(
    transformers=[
        ('num', numerical_transformer, numerical_features),
        ('cat', categorical_transformer, categorical_features)
    ],
    remainder='passthrough' # Keep other columns not specified
)

# Apply preprocessing to the features
X_preprocessed = preprocessor.fit_transform(df_social_media_cleaned)

print("\nMissing values handled and categorical features encoded.")
print(f"Shape of preprocessed dataset: {X_preprocessed.shape}")

print("\nCleaned and preprocessed dataset ready for engagement prediction.")
```

```
Duplicate user profiles removed based on all columns (no 'Username' found).
Number of duplicate records removed: 0
DataFrame shape after removing duplicates: (500, 5)

Missing values before handling and encoding:
```

| | 0 |
|---|---|
| N | 0 |
| PROFILE | 0 |
| FOLLOWERS | 0 |
| POSTS | 100 |
| Platform | 0 |

```
dtype: int64

Missing values handled and categorical features encoded.
Shape of preprocessed dataset: (500, 473)

Cleaned and preprocessed dataset ready for engagement prediction.
```

Task 11 – Weather Dataset Feature Engineering

Task:

Use AI to preprocess a weather dataset for forecasting.

Instructions:

• Convert date column into datetime format.

• Handle missing values in temperature, humidity.

• Create new features (season, week_number).

• Normalize rainfall values.

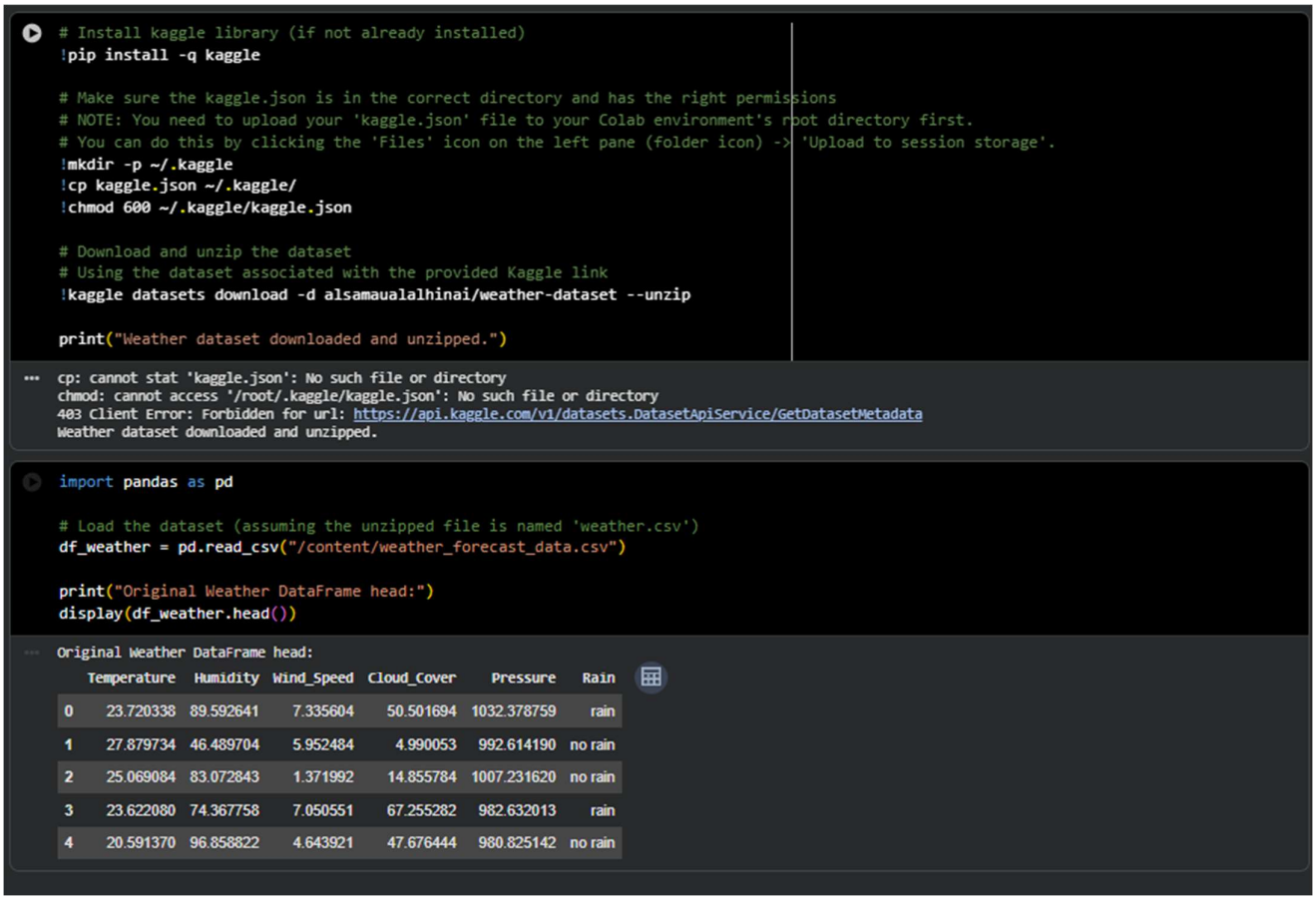
Dataset link:

https://www.kaggle.com/code/alsamaualalhinai/weather-dataset/input

Expected Output:

• A processed dataset ready for forecasting models.

screenshots:



```
# Install kaggle library (if not already installed)
!pip install -q kaggle

# Make sure the kaggle.json is in the correct directory and has the right permissions
# NOTE: You need to upload your 'kaggle.json' file to your Colab environment's root directory first.
# You can do this by clicking the 'Files' icon on the left pane (folder icon) -> 'Upload to session storage'.
!mkdir -p ~/.kaggle
!cp kaggle.json ~/.kaggle/
!chmod 600 ~/.kaggle/kaggle.json

# Download and unzip the dataset
# Using the dataset associated with the provided Kaggle link
!kaggle datasets download -d alsamaualalhinai/weather-dataset --unzip

print("Weather dataset downloaded and unzipped.")
```

```
cp: cannot stat 'kaggle.json': No such file or directory
chmod: cannot access '/root/.kaggle/kaggle.json': No such file or directory
403 Client Error: Forbidden for url: https://api.kaggle.com/v1/datasets.DatasetApiService/GetDatasetMetadata
Weather dataset downloaded and unzipped.
```

```
import pandas as pd

# Load the dataset (assuming the unzipped file is named 'weather.csv')
df_weather = pd.read_csv("/content/weather_forecast_data.csv")

print("Original Weather DataFrame head:")
display(df_weather.head())
```

```
Original Weather DataFrame head:
```

| | Temperature | Humidity | Wind_Speed | Cloud_Cover | Pressure | Rain |
|---|---|---|---|---|---|---|
| 0 | 23.720338 | 89.592641 | 7.335604 | 50.501694 | 1032.378759 | rain |
| 1 | 27.879734 | 46.489704 | 5.952484 | 4.990053 | 992.614190 | no rain |
| 2 | 25.069084 | 83.072843 | 1.371992 | 14.855784 | 1007.231620 | no rain |
| 3 | 23.622080 | 74.367758 | 7.050551 | 67.255282 | 982.632013 | rain |
| 4 | 20.591370 | 96.858822 | 4.643921 | 47.676444 | 980.825142 | no rain |

—

```python
from sklearn.preprocessing import StandardScaler
from sklearn.impute import SimpleImputer
import numpy as np

# Make a copy to avoid SettingWithCopyWarning
df_weather_cleaned = df_weather.copy()

# 1. Convert date column into datetime format
# Assuming the date column is named 'Date'
if 'Date' in df_weather_cleaned.columns:
    df_weather_cleaned['Date'] = pd.to_datetime(df_weather_cleaned['Date'], errors='coerce')
    # Drop rows where Date conversion failed (if any)
    df_weather_cleaned.dropna(subset=['Date'], inplace=True)
    print("\n'Date' column converted to datetime format.")
else:
    print("\n'Date' column not found. Please verify the date column name.")

# 2. Handle missing values in temperature, humidity
print("\nMissing values before handling:")
display(df_weather_cleaned[['Temperature', 'Humidity']].isnull().sum())

imputer = SimpleImputer(strategy='median')
df_weather_cleaned['Temperature'] = imputer.fit_transform(df_weather_cleaned[['Temperature']])
df_weather_cleaned['Humidity'] = imputer.fit_transform(df_weather_cleaned[['Humidity']])

print("\nMissing values after handling:")
display(df_weather_cleaned[['Temperature', 'Humidity']].isnull().sum())

# 3. Create new features (season, week_number)
# Season mapping (Northern Hemisphere)
def get_season(month):
    if month in [12, 1, 2]:
        return 'Winter'
    elif month in [3, 4, 5]:
        return 'Spring'
    elif month in [6, 7, 8]:
        return 'Summer'
    else:
        return 'Autumn'

if 'Date' in df_weather_cleaned.columns:
    df_weather_cleaned['Month'] = df_weather_cleaned['Date'].dt.month
    df_weather_cleaned['Season'] = df_weather_cleaned['Month'].apply(get_season)
    df_weather_cleaned['Week_Number'] = df_weather_cleaned['Date'].dt.isocalendar().week.astype(int)
    print("\nNew features 'Season' and 'Week_Number' created.")
else:
    print("\nDate column not available to create 'Season' and 'Week_Number'.")

# 4. Normalize rainfall values
# Assuming 'Rainfall' is the column name for rainfall values
if 'Rainfall' in df_weather_cleaned.columns:
    # Impute any remaining missing values in Rainfall before scaling, if necessary
    df_weather_cleaned['Rainfall'].fillna(df_weather_cleaned['Rainfall'].median(), inplace=True)

    scaler = StandardScaler()
    df_weather_cleaned['Rainfall_Normalized'] = scaler.fit_transform(df_weather_cleaned[['Rainfall']])
    print("\n'Rainfall' values normalized.")
else:
    print("\n'Rainfall' column not found. Please verify the rainfall column name.")

print("\nProcessed dataset head with new features and normalized values:")
display(df_weather_cleaned.head())
```

```
'Date' column not found. Please verify the date column name.

Missing values before handling:
              0
Temperature   0
   Humidity   0

dtype: int64

Missing values after handling:
              0
Temperature   0
   Humidity   0

dtype: int64

Date column not available to create 'Season' and 'Week_Number'.

'Rainfall' column not found. Please verify the rainfall column name.

Processed dataset head with new features and normalized values:
```

| | Temperature | Humidity | Wind_Speed | Cloud_Cover | Pressure | Rain |
|---|---|---|---|---|---|---|
| 0 | 23.720338 | 89.592641 | 7.335604 | 50.501694 | 1032.378759 | rain |
| 1 | 27.879734 | 46.489704 | 5.952484 | 4.990053 | 992.614190 | no rain |
| 2 | 25.069084 | 83.072843 | 1.371992 | 14.855784 | 1007.231620 | no rain |
| 3 | 23.622080 | 74.367758 | 7.050551 | 67.255282 | 982.632013 | rain |
| 4 | 20.591370 | 96.858822 | 4.643921 | 47.676444 | 980.825142 | no rain |

Task 12 – Movie Ratings Dataset Preparation

Task:

Clean and preprocess a movie ratings dataset using AI-generated scripts.

Instructions:

• Handle missing values in rating, genre.

• Encode categorical variables like genre, language.

• Remove duplicate movie entries.

• Scale rating values between 0 and 1.

Dataset link:

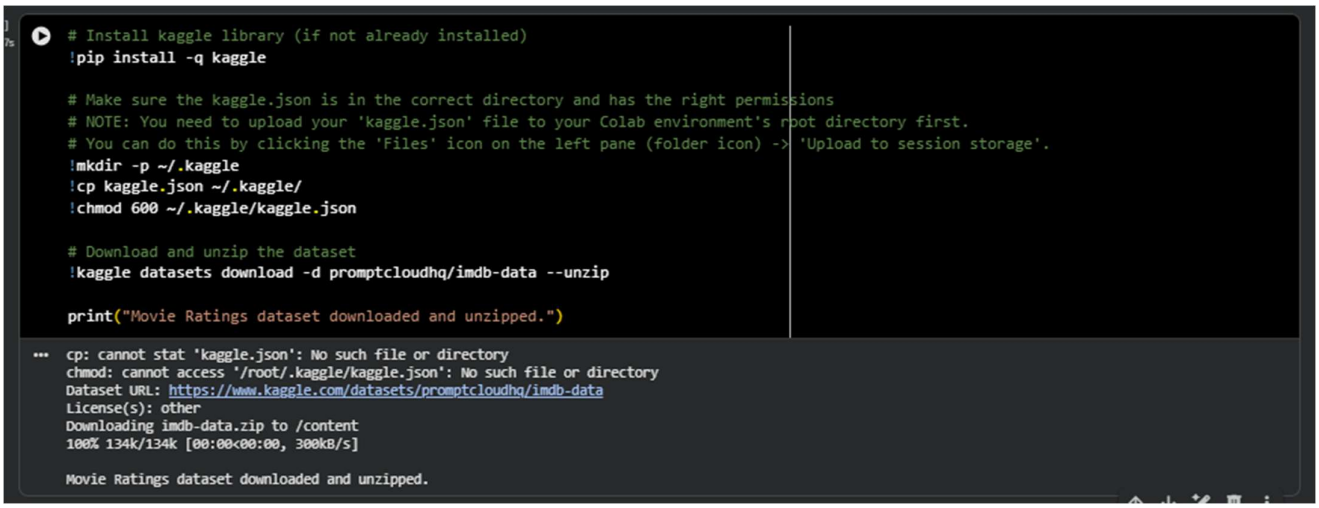
https://www.kaggle.com/datasets/PromptCloudHQ/imdb-

data

Expected Output:

• A dataset suitable for recommendation systems.

Screenshots:




—

```python
import pandas as pd

# Load the dataset (assuming the unzipped file is named 'IMDB-Movie-Data.csv')
df_movies = pd.read_csv("/content/IMDB-Movie-Data.csv")

print("Original Movie Ratings DataFrame head:")
display(df_movies.head())

print("\nDataFrame Info:")
df_movies.info()

print("\nMissing values before handling:")
display(df_movies.isnull().sum())
```

```
Original Movie Ratings DataFrame head:
```

| | Rank | Title | Genre | Description | Director | Actors | Year | Runtime (Minutes) | Rating | Votes | Revenue (Millions) | Metascore |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 1 | Guardians of the Galaxy | Action,Adventure,Sci-Fi | A group of intergalactic criminals are forced ... | James Gunn | Chris Pratt, Vin Diesel, Bradley Cooper, Zoe S... | 2014 | 121 | 8.1 | 757074 | 333.13 | 76.0 |
| 1 | 2 | Prometheus | Adventure,Mystery,Sci-Fi | Following clues to the origin of mankind, a te... | Ridley Scott | Noomi Rapace, Logan Marshall-Green, Michael Fa... | 2012 | 124 | 7.0 | 485820 | 126.46 | 65.0 |
| 2 | 3 | Split | Horror,Thriller | Three girls are kidnapped by a man with a diag... | M. Night Shyamalan | James McAvoy, Anya Taylor-Joy, Haley Lu Richar... | 2016 | 117 | 7.3 | 157606 | 138.12 | 62.0 |
| 3 | 4 | Sing | Animation,Comedy,Family | In a city of humanoid animals, a hustling thea... | Christophe Lourdelet | Matthew McConaughey,Reese Witherspoon, Seth Ma... | 2016 | 108 | 7.2 | 60545 | 270.32 | 59.0 |
| 4 | 5 | Suicide Squad | Action,Adventure,Fantasy | A secret government agency recruits some of th... | David Ayer | Will Smith, Jared Leto, Margot Robbie, Viola D... | 2016 | 123 | 6.2 | 393727 | 325.02 | 40.0 |

```
DataFrame Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 12 columns):
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
 0   Rank                1000 non-null   int64
 1   Title               1000 non-null   object
 2   Genre               1000 non-null   object
 3   Description         1000 non-null   object
 4   Director            1000 non-null   object
 5   Actors              1000 non-null   object
 6   Year                1000 non-null   int64
 7   Runtime (Minutes)   1000 non-null   int64
 8   Rating              1000 non-null   float64
 9   Votes               1000 non-null   int64
 10  Revenue (Millions)  872 non-null    float64
 11  Metascore           936 non-null    float64
dtypes: float64(3), int64(4), object(5)
memory usage: 93.9+ KB

Missing values before handling:
```

| | 0 |
|---|---|
| Rank | 0 |
| Title | 0 |
| Genre | 0 |
| Description | 0 |
| Director | 0 |
| Actors | 0 |
| Year | 0 |
| Runtime (Minutes) | 0 |
| Rating | 0 |
| Votes | 0 |
| Revenue (Millions) | 128 |

```python
# Make a copy to avoid SettingWithCopyWarning
df_movies_cleaned = df_movies.copy()

# 1. Handle missing values in 'Rating' and 'Genre'
print("\nHandling missing values...")

# Impute 'Rating' with the median
df_movies_cleaned['Rating'].fillna(df_movies_cleaned['Rating'].median(), inplace=True)

# Impute 'Genre' with the mode
df_movies_cleaned['Genre'].fillna(df_movies_cleaned['Genre'].mode()[0], inplace=True)

print("\nMissing values after handling:")
display(df_movies_cleaned[['Rating', 'Genre']].isnull().sum())
```

```
Handling missing values...

Missing values after handling:
/tmp/ipykernel_3362/1409206145.py:8: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.


  df_movies_cleaned['Rating'].fillna(df_movies_cleaned['Rating'].median(), inplace=True)
/tmp/ipykernel_3362/1409206145.py:11: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.


  df_movies_cleaned['Genre'].fillna(df_movies_cleaned['Genre'].mode()[0], inplace=True)
```

| | 0 |
|---|---|
| Rating | 0 |
| Genre | 0 |

```
dtype: int64
```

```
# One-Hot Encoding for 'Genre'
encoder = OneHotEncoder(handle_unknown='ignore', sparse_output=False)

# Split 'Genre' column by comma and expand to multiple rows for multi-genre movies
df_movies_cleaned['Genre_List'] = df_movies_cleaned['Genre'].apply(lambda x: x.split(','))
exploded_genres = df_movies_cleaned.explode('Genre_List')
exploded_genres['Genre_List'] = exploded_genres['Genre_List'].str.strip()

# Fit and transform the 'Genre_List' for one-hot encoding
encoded_genres = encoder.fit_transform(exploded_genres[['Genre_List']])
encoded_genre_df = pd.DataFrame(encoded_genres, columns=encoder.get_feature_names_out(['Genre_List']))

# Group back by original movie index and sum the one-hot encoded columns
df_genres_encoded = encoded_genre_df.groupby(exploded_genres.index).sum()

# Concatenate with the cleaned dataframe and drop the original 'Genre' and 'Genre_List' columns
df_movies_cleaned = pd.concat([df_movies_cleaned.drop(columns=['Genre', 'Genre_List']), df_genres_encoded], axis=1)

# If a 'Language' column existed, it would be handled similarly:
# if 'Language' in df_movies_cleaned.columns:
#     language_encoder = OneHotEncoder(handle_unknown='ignore', sparse_output=False)
#     encoded_languages = language_encoder.fit_transform(df_movies_cleaned[['Language']])
#     encoded_language_df = pd.DataFrame(encoded_languages, columns=language_encoder.get_feature_names_out(['Language']))
#     df_movies_cleaned = pd.concat([df_movies_cleaned.drop(columns=['Language']), encoded_language_df], axis=1)

print("Categorical variables encoded.")
display(df_movies_cleaned.head())
```

```
Encoding categorical variables...
Categorical variables encoded.
```

| | Rank | Title | Description | Director | Actors | Year | Runtime (Minutes) | Rating | Votes | Revenue (Millions) | ... | Genre_List_Horror | Genre_List_Music | Genre_List_Musical | Genre_List_Mystery | Genre_List_Romance | Genre_List_Sci-Fi | Genre_List_Sport | Genre_List_Thri |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 1 | Guardians of the Galaxy | A group of intergalactic criminals are forced ... | James Gunn | Chris Pratt, Vin Diesel, Bradley Cooper, Zoe S... | 2014 | 121 | 8.1 | 757074 | 333.13 | ... | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 1.0 | 0.0 | |
| 1 | 2 | Prometheus | Following clues to the origin of mankind, a te... | Ridley Scott | Noomi Rapace, Logan Marshall-Green, Michael Fa... | 2012 | 124 | 7.0 | 485820 | 126.46 | ... | 0.0 | 0.0 | 0.0 | 1.0 | 0.0 | 1.0 | 0.0 | |
| 2 | 3 | Split | Three girls are kidnapped by a man with a diag... | M. Night Shyamalan | James McAvoy, Anya Taylor-Joy, Haley Lu Richar... | 2016 | 117 | 7.3 | 157606 | 138.12 | ... | 1.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | |
| 3 | 4 | Sing | In a city of humanoid animals, a hustling thea... | Christophe Lourdelet | Matthew McConaughey,Reese Witherspoon, Seth Ma... | 2016 | 108 | 7.2 | 60545 | 270.32 | ... | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | |
| 4 | 5 | Suicide Squad | A secret government agency recruits some of th... | David Ayer | Will Smith, Jared Leto, Margot Robbie, Viola D... | 2016 | 123 | 6.2 | 393727 | 325.02 | ... | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | |

5 rows × 31 columns

```
from sklearn.preprocessing import MinMaxScaler

# 4. Scale 'Rating' values between 0 and 1
print("\nScaling 'Rating' values...")

scaler = MinMaxScaler()
df_movies_cleaned['Rating_Scaled'] = scaler.fit_transform(df_movies_cleaned[['Rating']])

print("\n'Rating' column scaled to [0, 1].")

print("\nCleaned and preprocessed DataFrame head:")
display(df_movies_cleaned.head())
```

```
Scaling 'Rating' values...

'Rating' column scaled to [0, 1].

Cleaned and preprocessed DataFrame head:
```

| | Rank | Title | Description | Director | Actors | Year | Runtime (Minutes) | Rating | Votes | Revenue (Millions) | ... | Genre_List_Music | Genre_List_Musical | Genre_List_Mystery | Genre_List_Romance | Genre_List_Sci-Fi | Genre_List_Sport | Genre_List_Thriller | Genre_List_Wa |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 1 | Guardians of the Galaxy | A group of intergalactic criminals are forced ... | James Gunn | Chris Pratt, Vin Diesel, Bradley Cooper, Zoe S... | 2014 | 121 | 8.1 | 757074 | 333.13 | ... | 0.0 | 0.0 | 0.0 | 0.0 | 1.0 | 0.0 | 0.0 | 0 |
| 1 | 2 | Prometheus | Following clues to the origin of mankind, a te... | Ridley Scott | Noomi Rapace, Logan Marshall-Green, Michael Fa... | 2012 | 124 | 7.0 | 485820 | 126.46 | ... | 0.0 | 0.0 | 1.0 | 0.0 | 1.0 | 0.0 | 0.0 | 0 |
| 2 | 3 | Split | Three girls are kidnapped by a man with a diag... | M. Night Shyamalan | James McAvoy, Anya Taylor-Joy, Haley Lu Richar... | 2016 | 117 | 7.3 | 157606 | 138.12 | ... | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 1.0 | 0 |
| 3 | 4 | Sing | In a city of humanoid animals, a hustling thea... | Christophe Lourdelet | Matthew McConaughey,Reese Witherspoon, Seth Ma... | 2016 | 108 | 7.2 | 60545 | 270.32 | ... | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0 |
| 4 | 5 | Suicide Squad | A secret government agency recruits some of th... | David Ayer | Will Smith, Jared Leto, Margot Robbie, Viola D... | 2016 | 123 | 6.2 | 393727 | 325.02 | ... | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0 |

5 rows × 32 columns